

CHAPTER OUTLINE

INDIVIDUAL CLASSES OF DIURETICS, 1709
ADAPTATION TO DIURETIC THERAPY, 1721

CLINICAL USES OF DIURETICS, 1725
ADVERSE EFFECTS OF DIURETICS, 1734

KEY POINTS

- Dietary sodium restriction, selection of a diuretic with a prolonged action, or more frequent administration of the diuretic will enhance NaCl loss by limiting postdiuretic salt retention.
- During prolonged diuretic administration, and in the case of diuretic resistance, subjects may be particularly responsive to another class of diuretic.
- Torsemide is the preferred loop diuretic in heart failure because of better bioavailability, absence of hypokalemia, and superior outcomes compared with other loop diuretics.
- In heart failure, diuretic responsiveness is impaired, as shown by a shift to the right in the natriuresis-excretion relationship of diuretics.
- Compensatory distal reabsorption rather than reduced delivery drives diuretic resistance in heart failure.
- In chronic kidney disease, the maximal increase in fractional excretion of Na⁺ produced by loop diuretics is maintained, but the absolute response is limited.
- The addition of an SGLT2 inhibitor may help relieve loop diuretic resistance in patients with diabetes mellitus while providing protection from progressive loss of the GFR.
- The risk of thiazide-induced hyponatremia increases in subjects with a variant in *SLC02A1*, which inactivates a prostaglandin transporter and allows prostaglandin E₂ to stimulate EP₄ receptors and insert aquaporin-2 water channels.

This chapter reviews the mechanisms of actions, physiologic adaptations, clinical uses, and adverse effects of diuretics. The term *diuretics* refers to several groups of drugs that share the effect of increasing urinary output. Most diuretics target sodium transport proteins in the kidney (natriuretics). Novel targets for which drugs are in preclinical development also include potassium and urea channels (urearetics; see “[Novel Diuretics in Preclinical Development](#)”). Other targets of diuretics include enzymes (e.g., carbonic anhydrase, nephrilysin) or receptors (e.g., mineralocorticoid receptors) that indirectly control renal sodium reabsorption. Vasopressin receptor antagonists (vaptans) increase diuresis by selectively inhibiting water reabsorption (aquaretics). Osmotic diuretics are filtered but not reabsorbed and create a reverse osmotic gradient that inhibits the reabsorption of water and, indirectly, NaCl (solvent drag). Other diuretics also rely on osmotic diuresis, including the novel sodium-glucose transporter type 2 (SGLT2) inhibitors and the urea channel inhibitors.

The major sodium transport targets for diuretic drugs have been defined and their genes cloned. The effects of disease on diuretic kinetics are discussed in this chapter because they predict the required dosage modifications. Loop diuretics and thiazides are the most widely used diuretics, and the physiologic adaptations to their prolonged use are described. Diuretic resistance, its management, and major adverse effects of therapy are discussed. This discussion provides a framework for the design of strategies to maximize the desired actions while minimizing the unwanted effects. The chapter also includes a discussion of the practical use of diuretics in the treatment of specific clinical conditions.

Other chapters discuss the treatment of hypertension by diuretic drugs (Chapter 49), diuretic-induced changes in potassium excretion (Chapter 17), acid-base disturbances (Chapter 16), divalent cation excretion and nephrolithiasis (Chapters 18 and 38), the syndrome of inappropriate antidiuretic hormone (SIADH) secretion (Chapter 15), and

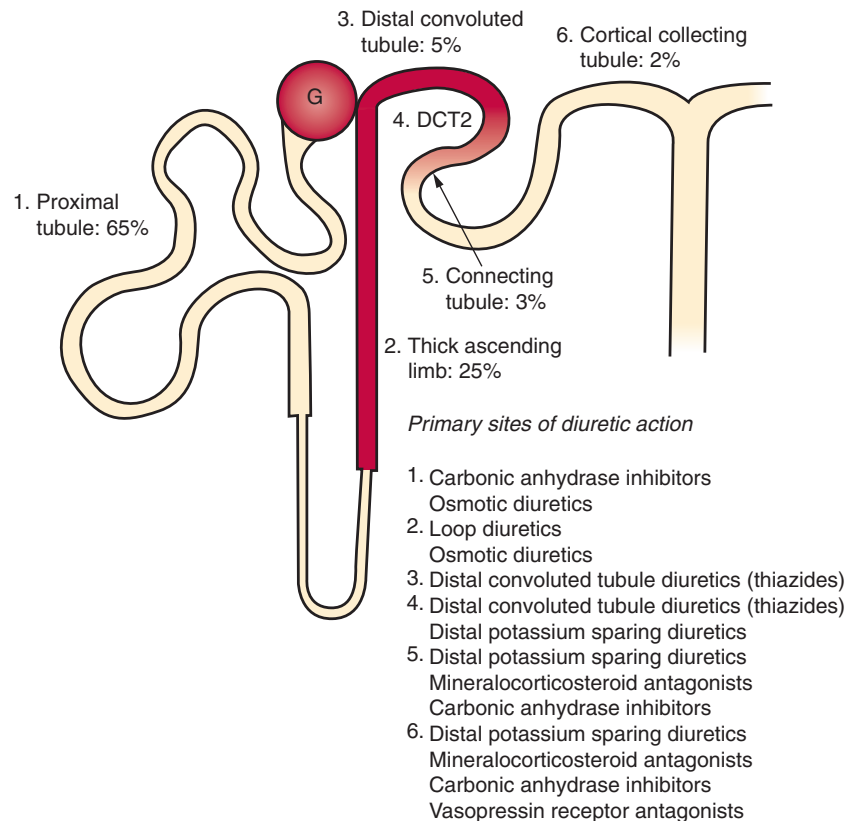


Fig. 50.1 Nephron diagram showing the primary sites of diuretic action and the approximate percentage of filtered sodium reabsorbed at each. DCT2, Late segment of distal convoluted tubule; G, glomerulus.

acute kidney injury ([AKI]; Chapter 28). Diuretics have been reviewed extensively.¹⁻⁴ More extensive and historical references appeared in previous editions of this chapter; the interested reader is referred to editions 7 to 10 of *Brenner & Rector's The Kidney* for more detailed references.

INDIVIDUAL CLASSES OF DIURETICS

The major sites of action of diuretics and fractions of filtered Na^+ reabsorbed at the corresponding nephron segments are summarized in Fig. 50.1.

CARBONIC ANHYDRASE INHIBITORS

SITES AND MECHANISMS OF ACTION

In the kidney, carbonic anhydrase inhibitors (CAIs) act primarily on proximal tubule cells to inhibit bicarbonate absorption (Fig. 50.2). An additional, more modest, effect along the distal nephron, however, is also observed.⁵ Carbonic anhydrase (CA), a metalloenzyme containing one zinc atom per molecule, is important in sodium bicarbonate reabsorption and hydrogen ion secretion by renal epithelial cells. The biochemical, morphologic, and functional properties of carbonic anhydrase have been reviewed.^{6,7}

CA is expressed by many tissues, including erythrocytes, kidney, gut, ciliary body, choroid plexus, and glial cells. Although at least 14 isoforms of CA have been identified, two play predominant roles in renal acid-base homeostasis, CA II and CA IV. CA II is widely expressed, comprising the

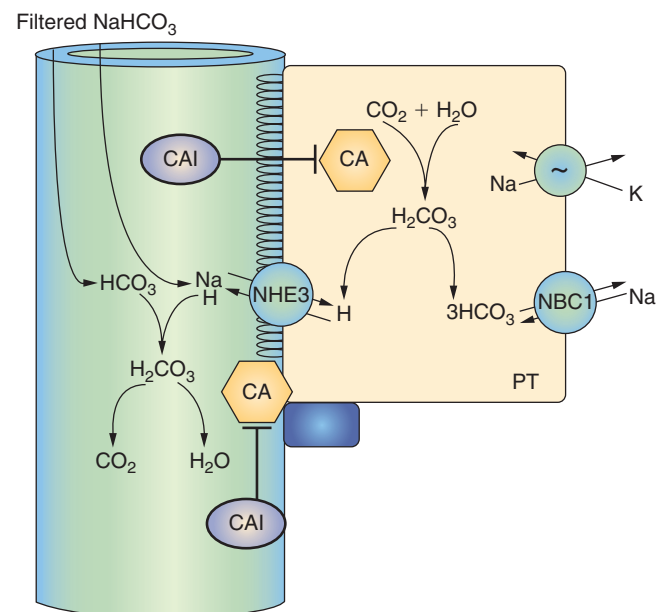


Fig. 50.2 Mechanisms of diuretic action in the proximal tubule. The figure shows a functional model of proximal tubule cells; many transport proteins are omitted from the model for clarity. Inside the cell, carbonic anhydrase (CA) catalyzes the formation of HCO_3^- from OH^- and CO_2 . Bicarbonate leaves the cell via the sodium-bicarbonate transporter (NBC1). A second pool of carbonic anhydrase is located in the brush border. This participates in disposing of the carbonic acid formed from filtered bicarbonate and secreted H^+ . Both pools of carbonic anhydrase are inhibited by acetazolamide and other carbonic anhydrase inhibitors (CAIs; see text for details). NBC1, Sodium-bicarbonate cotransporter 1.

enzyme expressed by red blood cells and a variety of secretory and absorptive epithelia. In the kidney, CA II is expressed in the cytoplasm and accounts for 95% of renal CA.⁷ It is present in proximal tubule cells and intercalated cells of the aldosterone-sensitive distal nephron (ASDN).⁷ CA IV is expressed at the luminal border of the cells of the proximal thick ascending limb (TAL) of the loop of Henle and α -intercalated cells of the ASDN.⁸

The prototypic CAI is acetazolamide. However, many diuretics have some CAI action.⁹ This characteristic contributes to the weak inhibition of proximal reabsorption by furosemide and chlorothiazide and to the relaxation of vascular smooth muscle cells by high-dose furosemide.⁹

CAIs block the catalytic dehydration of luminal carbonic acid at the brush border of the proximal tubule, decrease the intracellular generation of H^+ required for countertransport with Na^+ , and decrease the peritubular capillary fluid uptake.¹⁰ CAIs also are also weak inhibitors of reabsorption in the TAL,¹¹ but the natriuretic efficacy of CAIs and loop diuretics (see later) is additive, confirming their independent mechanisms of action.¹² CAIs also inhibit bicarbonate reabsorption along the distal tubule, presumably by interfering with the action of α -intercalated cells.⁷ The first administration of a CAI causes a brisk alkaline diuresis. The excretion of Na^+ , K^+ , HCO_3^- , and PO_4^{2-} increases, whereas the excretion of titratable acid and NH_4^+ decrease sharply. Excretion of Ca^{2+} remains essentially unchanged. There is substantial kaliuresis owing to the presence of nonreabsorbable HCO_3^- and high flow rates in the distal nephron. However, hypokalemia is uncommon, because acidosis partitions K^+ out of cells.

Long-term CAI administration causes only a modest natriuresis, despite the magnitude of CA-dependent proximal Na^+ reabsorption. Several factors account for this:

1. CA is required for the reabsorption of HCO_3^- , whereas about two thirds of the proximal Na^+ reabsorption is accompanied by Cl^- .
2. Some proximal HCO_3^- reabsorption persists, even after apparently full inhibition of CA.¹³
3. Some of the HCO_3^- that is delivered out of the proximal tubule can be reabsorbed at more distal sites.¹³
4. The metabolic acidosis that develops limits the filtered load of HCO_3^- and thereby curtails the natriuresis.
5. The increased delivery of filtered Na^+ to the macula densa elicits a tubuloglomerular feedback (TGF)-induced reduction in the glomerular filtration rate (GFR).¹⁴

Micropuncture studies of mice with deletion of the proximal Na^+-H^+ exchanger, NHE3, have shown that inhibition of proximal Na reabsorption is largely balanced by reduced GFR,¹⁵ supporting this mechanism. Although CAIs only cause a modest natriuresis, the combined use of a thiazide and acetazolamide results in a brisk natriuresis.¹⁶ This observation may be explained by the fact that acetazolamide inhibits the HCO_3^-/Cl^- exchanger pendrin, which is upregulated secondary to upstream sodium chloride cotransporter (NCC) inhibition by thiazide diuretics.^{16,17}

PHARMACOKINETICS

Acetazolamide (Diamox) is readily absorbed. It is eliminated with a half-life ($t_{1/2}$) of 13 hours by tubular secretion, which is diminished during hypoalbuminemia.¹⁸ Methazolamide

(Neptazane) has less plasma protein binding, a longer $t_{1/2}$, and greater lipid solubility, all of which favor penetration into aqueous humor and cerebrospinal fluid. This agent has less renal effect and therefore was used to treat glaucoma.

CLINICAL INDICATIONS

The use of CAIs as diuretics is limited by their transient action, the development of metabolic acidosis, and a spectrum of adverse effects. They can be used with $NaHCO_3$ infusion to initiate an alkaline diuresis that increases the excretion of weakly acidic drugs (e.g., salicylates, phenobarbital) or acidic metabolites (e.g., urate). Chloride depletion metabolic alkalosis is best treated by the administration of Cl^- with K^+ or Na^+ . However, if this produces unacceptable extracellular volume (ECV) expansion, acetazolamide (250–500 mg/day) and KCl can be used to increase HCO_3^- excretion.

Metabolic alkalosis due to loop diuretics or thiazides can depress respiration in patients with chronic respiratory acidosis—for example, due to chronic obstructive pulmonary disease. This effect provides the rationale for use of a CAI. Indeed, the administration of acetazolamide to such subjects can reduce the arterial partial pressure of arterial carbon dioxide ($Paco_2$) and improve the partial pressure of oxygen (Pao_2). Because both $Paco_2$ and the plasma bicarbonate concentration ($PHCO_3$) decrease, there is little change in blood pH.¹⁹ However, a reduction in $PHCO_3$ limits the buffering capacity of blood. CAIs can increase the $Paco_2$ during metabolic acidosis or exercise, perhaps by depressing hypoxic ventilatory drive²⁰ and hypoxic pulmonary vasoconstriction,²¹ and can cause ventilation-perfusion imbalance.²² Nevertheless, acetazolamide (250 mg bid) can improve blood gas parameters in patients with chronic obstructive pulmonary disease.²³ Careful surveillance is required when CAIs are administered to such patients.

When used to treat glaucoma, CAIs diminish the transport of HCO_3^- and Na^+ by the ciliary process, thereby reducing the intraocular pressure.²⁴ CAIs also limit the formation of cerebrospinal fluid²⁵ and endolymph.²⁶ Currently, their use to treat glaucoma is typically limited to topical preparations, most commonly as combination agents. On the other hand, oral acetazolamide is still widely used as one of the therapeutic interventions in the management of idiopathic intracranial hypertension (IIH; also sometimes referred to as *pseudotumor cerebri*).²⁷

Acute mountain sickness is characterized by headache, nausea, drowsiness, insomnia, shortness of breath, dizziness, and malaise after an abrupt ascent. Acetazolamide is useful in dosages of 250 to 750 mg daily as prophylaxis against mountain sickness, probably through stimulating respiration and diminishing cerebral blood flow and cerebrospinal fluid formation.^{28–30} Used in established mountain sickness, acetazolamide improves oxygenation and pulmonary gas exchange.³¹ It can stimulate ventilation in patients with central sleep apnea.³² In patients with IIH, acetazolamide administration, along with a low-sodium weight reduction diet, modestly improves visual field function.³³

CAIs are effective in the prophylaxis of hypokalemic periodic paralysis because they diminish the influx of K^+ into cells.³⁴ Paradoxically, they are also useful in the treatment of hyperkalemic periodic paralysis.³⁵

It was recently shown, in mice, that acetazolamide attenuates lithium-induced nephrogenic diabetes insipidus, with fewer

adverse effects than thiazides and amiloride.³⁶ This effect was ascribed to a reduction in the GFR and activation of TGF, but may also indicate a specific effect in the collecting duct. Clinical data are needed to assess the corresponding effect in humans.

Other indications for CAIs are experimental. They include possible applications in diseases as diverse as obesity, cancer, and infection.³⁷

ADVERSE EFFECTS

Patients taking CAIs may complain of weakness, lethargy, abnormal taste, paresthesia, gastrointestinal distress, malaise, and decreased libido. These symptoms can be diminished by NaHCO_3 but this agent increases the risk of nephrocalcinosis and nephrolithiasis.³⁸ Overall, symptomatic metabolic acidosis develops in 50% of patients with glaucoma treated with CAIs.³⁹

Older patients or those with diabetes mellitus or chronic kidney disease (CKD) can experience a serious metabolic acidosis if given a CAI.³⁹ An alkaline urine favors partitioning of renal ammonia into blood rather than its elimination in urine. An increase in blood ammonia may precipitate encephalopathy in patients with liver failure.³⁹

Acetazolamide increases the risk of nephrolithiasis by more than tenfold.⁴⁰ CAIs occasionally cause allergic reactions, hepatitis, and blood dyscrasias.⁴¹ They can cause osteomalacia when used with phenytoin or phenobarbital.⁴²

OSMOTIC DIURETICS

SITES AND MECHANISMS OF ACTION

Osmotic diuretics are substances that are freely filtered but poorly reabsorbed.⁴³ Mannitol is the prototypic osmotic diuretic, although sorbitol and glycerol have similar actions. In the water-permeable nephron segments of the proximal nephron and thin limbs of the loop of Henle, fluid reabsorption concentrates filtered mannitol sufficiently to diminish tubular fluid reabsorption. Ongoing Na^+ reabsorption lowers the tubular fluid $[\text{Na}^+]$ and creates a gradient for backflux of reabsorbed Na^+ into the tubule. Increased distal flow stimulates K^+ secretion.

Mannitol is a hypertonic solute that abstracts water from cells. The increase in total renal blood flow (RBF) relates in part to hemodilution and a decrease in hematocrit and viscosity. Mannitol increases the medullary blood flow and decreases the medullary solute gradient, thereby preventing urinary concentration. The rise in renal plasma flow and drop in plasma colloid osmotic pressure can increase the GFR.⁴⁴

PHARMACOKINETICS AND DOSAGE

Mannitol is distributed exclusively to the extracellular fluid. It is filtered freely at the glomerulus. Consequently, the $t_{1/2}$ for plasma clearance of mannitol depends on the GFR and is prolonged from 1 to 36 hours in those with advanced CKD.⁴⁵ It can be infused intravenously in daily doses of 50 to 200 g as a 15% or 20% solution or 1.5 to 2.0 g/kg of 20% mannitol over 30 to 60 minutes to treat elevated intraocular or intracranial pressure.⁴³

CLINICAL INDICATIONS

Mannitol has been evaluated for the prophylaxis of AKI, but controlled trials of its use in patients at risk for AKI have

not had positive results.^{46,47} The rationale for such trials includes mannitol's ability to expand the ECV, block TGF, maintain the GFR, increase RBF and tubule fluid flow, prevent tubule obstruction from shed cell constituents or crystals, reduce renal edema, redistribute blood flow from the outer cortex to the relatively hypoxic inner cortex and outer medulla, and scavenge oxygen radicals.^{43,44} It can protect against AKI in cadaveric kidney transplant recipients.⁴³ The use of diuretics to convert oliguric to nonoliguric AKI is discussed later (see "[Clinical Uses of Diuretics](#)").

A trial of mannitol therapy for cerebral edema complicating hepatic failure has demonstrated a markedly better survival of 47%, compared with only 6% in the control group.⁴⁸ Mannitol is recommended for the management of severe head injury.^{49,50} It is more effective than loop diuretics or hypertonic saline in reducing brain water content.⁵¹ Mannitol can reverse the dialysis disequilibrium syndrome.⁵²

ADVERSE EFFECTS

The effects of mannitol on plasma electrolyte concentrations are complex. The osmotic abstraction of cell water initially causes hypertonic hyponatremia and hypochloremia. Later, when the excess extracellular fluid (ECF) is excreted, the decrease in cell water concentrates K^+ and H^+ within cells, thereby increasing the gradient for their diffusion into the ECF and leading to hyperkalemic acidosis. Normally, these electrolyte changes are rapidly corrected by the kidney, provided that renal function is adequate. Later, hypernatremic dehydration may develop if free water is not provided because urinary concentrating ability is inhibited.⁵³

Expansion of ECV, hemodilution, and hyperkalemic metabolic acidosis occur in patients with renal failure who cannot eliminate the drug. Circulatory overload, pulmonary edema, central nervous system depression, and severe hyponatremia require urgent hemodialysis.⁵⁴ Doses of more than 200 g/day can cause renal vasoconstriction and AKI.⁴³

LOOP DIURETICS

SITES AND MECHANISMS OF ACTION

The primary action of loop diuretics occurs from the luminal aspect of the TAL ([Fig. 50.3](#)). An electroneutral Na-K-2Cl cotransporter, termed "NKCC2," is located at the luminal membrane.^{55,56} This cotransporter, a member of the solute carrier family 12 (gene symbol *SLC12A1*), mediates Na^+ and Cl^- movement across the cell. A high luminal K^+ conductance, via the renal outer medullary K^+ (ROMK) channel, allows most of the K^+ to recycle across the luminal membrane.⁵⁷ Coupled with the electrogenic exit of Cl^- across the basolateral membrane, the activity of the NKCC2 generates a transepithelial voltage, oriented positively with the lumen relative to interstitial fluid. The primary energy for transport across TAL cells is provided via the basolateral sodium pump, Na^+-K^+- adenosine triphosphatase (ATPase), which maintains a low intracellular $[\text{Na}^+]$. Additional details concerning mechanisms of solute reabsorption by TAL cells can be found in Chapter 6. Loop diuretics are organic anions that bind to the NKCC2 from the luminal surface. Early studies have shown that [^3H] bumetanide binds to membranes that express the NKCC proteins and that Cl^- competes for the same binding site on the transport protein.⁵⁸ Studies using chimeric NKCC molecules have investigated sites of bumetanide binding and

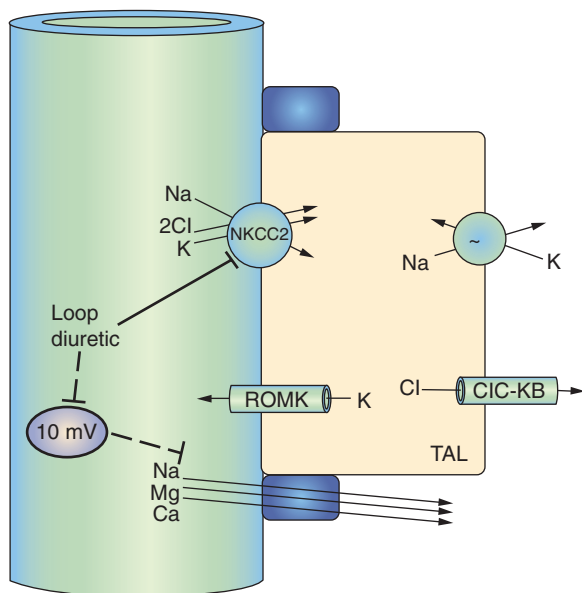


Fig. 50.3 Mechanisms of diuretic action along the loop of Henle. The figure shows a model of thick ascending limb cells. Na^+ and Cl^- are reabsorbed across the apical membrane via the loop diuretic-sensitive Na-K-2Cl cotransporter 2 (NKCC2). Loop diuretics bind to and block this pathway directly. Note that the transepithelial voltage along the thick ascending limb is oriented with the lumen positive relative to blood (circled value, given in millivolts [mV]). This transepithelial voltage drives a component of Na^+ (and calcium and magnesium; see Fig. 50.4) reabsorption via the paracellular pathway. This component of Na^+ absorption is also reduced by loop diuretics because they reduce the transepithelial voltage. ClC-KB, Chloride channel protein; ROMK, renal outer medullary K^+ channel; TAL, thick ascending limb.

interactions with ions by determining effects on the ion transport of heterologously expressed NKCC proteins; it was found that changes in amino acids that affect bumetanide binding are not the same as patterns of changes affecting the kinetics of ion translocation.^{59,60} Nevertheless, the second membrane-spanning segment of NKCC2 does appear to participate in both anion affinity and bumetanide affinity.^{59,60} A clearer picture of the details of diuretic and ion interaction with the NKCC protein must await structural biology advances for atomic level resolution.

NKCC2 is expressed on the apical membranes of medullary and cortical TALs and macula densa segments.^{61,62} Its abundance is increased by prolonged infusion of saline or furosemide.⁶¹ A closely related gene, *NKCC1*, encodes a protein that is widely expressed in transporting epithelia.⁵⁵ In contrast to NKCC2, *NKCC1* is implicated in the uptake and secretion of Cl^- and NH_4^+ at the basolateral membrane of the medullary CDs.⁶³

Hormones that stimulate cyclic adenosine monophosphate (cAMP), such as arginine vasopressin (AVP), enhance TAL reabsorption and should enhance the response to loop diuretics.⁶⁴ In contrast, those that stimulate cyclic guanosine monophosphate (cGMP), such as nitric oxide and atrial natriuretic peptide (ANP), those that increase intracellular $[\text{Ca}^{2+}]$, such as 20-hydroxyeicosatetraenoic acid (20-HETE), or those that activate the Ca^{2+} (polyvalent cation)-sensing protein⁶⁵ inhibit TAL reabsorption and reduce the response to loop diuretics.⁶⁶

The rat TAL also transports NH_4^+ ,⁶⁷ which can substitute for K^+ on NKCC2. In the rat, there is a luminal $\text{Na}^+\text{-H}^+$ countertransporter that contributes to tubular fluid acidification. Loop diuretics block the luminal entry of Na^+ via NKCC2, but not the peritubular exit via the $\text{Na}^+\text{-K}^+\text{-ATPase}$, and thereby reduce the intracellular $[\text{Na}^+]$ sufficiently to promote luminal Na^+ uptake via the $\text{Na}^+\text{-H}^+$ countertransport process. This is one reason that furosemide stimulates acid excretion in the rat.^{68,69} In some studies, furosemide has not affected net acid excretion or urine pH in normal human subjects.⁷⁰

Loop diuretics reduce proximal fluid reabsorption modestly. This effect has been ascribed to a weak CAI action. However, furosemide depresses proximal reabsorption in tubules perfused with HCO_3^- -free solutions.⁷¹ Moreover, bumetanide, which is a much less potent inhibitor of CA, also impairs proximal fluid reabsorption.⁷²

Furosemide exerts two contrasting effects on reabsorption in the superficial distal tubule. Increased delivery to the unsaturated distal tubule reabsorption process increases Na^+ reabsorption.⁶⁸ However, Velazquez and Wright perfused rat distal tubules *in vivo* to obviate the confounding effects of altered delivery.⁷³ They concluded that furosemide, but not bumetanide, was a weak inhibitor of the thiazide-sensitive NCC, although the cloned NCC from flounder was not sensitive to furosemide when expressed in *Xenopus laevis* oocytes.⁷⁴ Loop diuretics also inhibit NaCl transport in short descending limbs of the loop of Henle⁷⁵ and collecting ducts (CDs).⁷⁶ Although the TAL is clearly the major site of action of loop diuretics, actions at other nephron segments contribute to the natriuresis by blunting the expected increase in reabsorption in the proximal tubule (in response to volume depletion) and the distal nephron (in response to increased load). Reabsorption of solute from the water-impermeable TAL segments dilutes the tubular fluid and concentrates the interstitium. Its inhibition by loop diuretics impairs free water excretion during water loading and free water reabsorption during dehydration.⁷⁷

Loop diuretics increase the fractional excretion of Ca^{2+} by up to 30%.⁷⁸ The predominant mechanism is a decrease in the magnitude of the lumen-positive transepithelial potential (Fig. 50.4; see Fig. 50.3). A large fraction of transepithelial Ca^{2+} transport along the TAL traverses a paracellular pathway involving claudins 16 and 19 and is driven by the lumen-positive transepithelial potential.⁷⁹ By reducing the magnitude of this potential, loop diuretics lower passive calcium absorption along this segment. A second mechanism has been observed in some experiments, involving active Ca^{2+} transport, but this pathway is not affected by loop diuretics.⁸⁰

The loop of Henle is also a major nephron segment for the reabsorption of Mg^{2+} .⁷⁸ Mg^{2+} transport along the TAL, like Ca^{2+} transport, traverses a paracellular pathway that involves claudins and is driven by the transepithelial potential difference. Loop diuretics can increase fractional Mg^{2+} excretion by more than 60%⁷⁴ by diminishing voltage-dependent paracellular transport⁷⁸ (see Figs. 50.3 and 50.4 and see later discussion of adverse effects). Loop diuretics initially increase urate excretion by inhibiting proximal urate transport.⁸¹ However, there is a succeeding reduction in urate clearance that is largely secondary to volume depletion.⁸² The total RBF is maintained or increased and the GFR is little changed during the administration of loop diuretics to

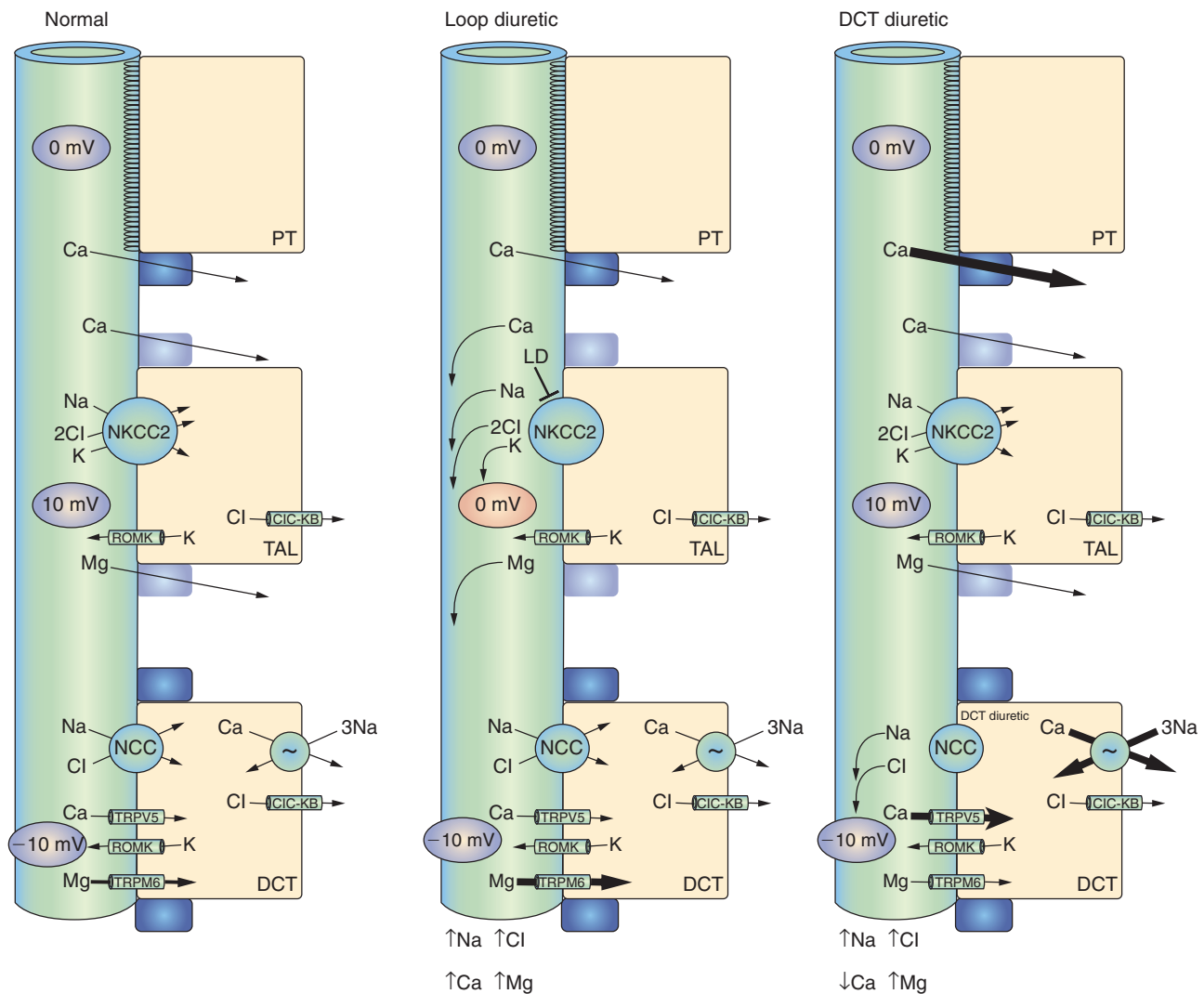


Fig. 50.4 Possible mechanisms of diuretic effects on calcium and magnesium excretion. Typical cells from the proximal tubule (PT), thick ascending limb (TAL), and distal convoluted tubule (DCT) are shown. Calcium reabsorption occurs along the distal convoluted tubule largely via a transient receptor potential channel (TRPV5). Magnesium reabsorption occurs along the distal convoluted tubule largely via a transient receptor potential channel (TRPM6). Transepithelial voltages (representative but arbitrary values, given in millivolts [mV]) are shown. Net effects on electrolyte excretion are shown at the bottom. Normal conditions are at the left. Treatment with loop diuretics (LDs) is shown in the middle; treatment with DCT diuretics is shown on the right. Loop diuretics reduce the magnitude of the lumen-positive transepithelial voltage, thereby retarding passive calcium and magnesium reabsorption. Passive calcium and magnesium reabsorption appears to traverse the paracellular pathway. Long-term treatment, especially with DCT diuretics, increases proximal Na^+ and Ca^{2+} reabsorption; thus, less calcium is delivered distally. Enhanced distal calcium absorption, driven by DCT diuretics, also occurs. Effects of DCT diuretics to increase magnesium excretion remain incompletely understood. CIC-KB, Chloride channel protein; NCC, Na-Cl transporter; NKCC2, Na-K-2Cl cotransporter 2; ROMK, renal outer medullary K^+ channel; $\uparrow$, increase(d); $\downarrow$, decrease(d).

normal subjects.⁸³ However, there is a marked redistribution of blood flow from the inner to the outer cortex.⁸⁴ The fall in papillary plasma flow depends on angiotensin II.⁸⁵ Furosemide increases the renal generation of prostaglandins.⁸⁶ Blockade of cyclooxygenase prevents furosemide-induced renal vasodilation.⁸⁷

The macula densa participates importantly both in renin secretion and in TGF-mediated control of GFR. NaCl entry into macula densa cells regulates both processes; thus, loop diuretics affect both TGF and renin secretion. When the luminal NaCl concentration at the macula densa rises, as during ECV expansion, NaCl entry into macula densa cells leads to the production of adenosine, which interacts with

adenosine 1 receptors on vascular smooth muscle and/or extraglomerular mesangial cells, activating phospholipase C. This activation leads to the depolarization and activation of voltage-dependent Ca^{2+} channels, which contract afferent arterioles and reduce the GFR (the TGF response).⁸⁸ NaCl transport across the luminal membrane of macula densa cells traverses the NKCC2.⁶² Loop diuretics, by blocking NaCl entry into macula densa cells, block TGF completely.⁸⁹ This is one reason that loop diuretics tend to preserve GFR despite ECV depletion.

Loop diuretics also stimulate renin secretion, both in the short term and long term. Although this effect results, in part, from ECV depletion, a major component is from direct

effects of loop diuretics on the macula densa. NaCl uptake into macula densa cells inhibits renin secretion acutely and inhibits renin synthesis chronically.⁹⁰ Macula densa cells were shown to express cyclooxygenase-2 (COX-2). Schnermann has shown that lowering NaCl concentration bathing a macula densa cell line acutely increases the release of prostaglandin E₂ (PGE₂), followed by a delayed induction of COX-2 expression.⁹⁰ A similar stimulation of COX-2 expression was also caused by furosemide and bumetanide. A lowering of the medium chloride concentration was followed by rapid phosphorylation of p44/42 and p38 MAP kinases, and the presence of p44/42 and p38 inhibitors prevented the stimulation of COX-2 expression by low chloride. In summary, a decrease in luminal NaCl concentration activates and transcriptionally induces COX-2, causing PGE₂ release, and PGE₂ receptor type 4 (EP4)-mediated stimulation of renin secretion and renin synthesis. Nitric oxide synthase inhibition with L-NAME (*N*^G-nitro-L-arginine methyl ester) has been found to completely block the increase in renin messenger RNA (mRNA) after the administration of furosemide for 4 days by minipump infusion.⁹¹ Similarly, the increase in renin content in renal microvessels caused by a 5-day furosemide treatment was completely prevented by L-NAME.⁹² However, mice made deficient in both the neuronal and endothelial forms of nitric oxide synthase display relatively normal renin responses to loop diuretics.⁹³ These data have been interpreted to suggest that nitric oxide synthesis plays a permissive role in macula densa-mediated renin secretion.

PHARMACOKINETICS AND DIFFERENCES BETWEEN DRUGS

Loop diuretics are absorbed promptly after ingestion, but their bioavailabilities vary. Because bumetanide and torsemide are more completely absorbed than furosemide, changing from intravenous to oral dosing may require a doubling of the furosemide dose but does not require changing the bumetanide or torsemide dose. Moreover, there is considerable variation in furosemide absorption, both between patients and over time,⁹⁴ that is accentuated by food intake.^{95,96}

The highly variable bioavailability of furosemide compared with torsemide and the absence of hypokalemia with usual doses,⁹⁷ combined with the superior outcomes of torsemide in two clinical trials of patients with heart failure^{98,99} have prompted consideration that torsemide become the loop diuretic of choice, at least for heart failure.^{100–102} However, a class problem shared with all available loop diuretics is their short duration of action, 2 to 4 hours.⁹⁷ The resulting torrential diuresis (termed the *Niagara effect*) can be troubling, especially for patients with prostatism or stress incontinence. Moreover, the limited time of action of a daily dose of diuretic on the tubules provides some 20 hours for the kidneys to regain salt and water losses (see Fig. 50.10). An investigational extended-release formulation of torsemide that delivers torsemide to the circulation over 8 to 12 hours doubled salt and water losses in normal volunteers in the day after a single dose, without increasing potassium excretion.¹⁰³ Remarkably, one dose of the extended-release torsemide given to subjects ingesting 300 mmol of Na⁺ daily, caused a negative daily Na⁺ balance, whereas prior studies with furosemide at this level of salt intake reported no net loss of Na⁺ because of postdiuretic salt retention.¹⁰⁴

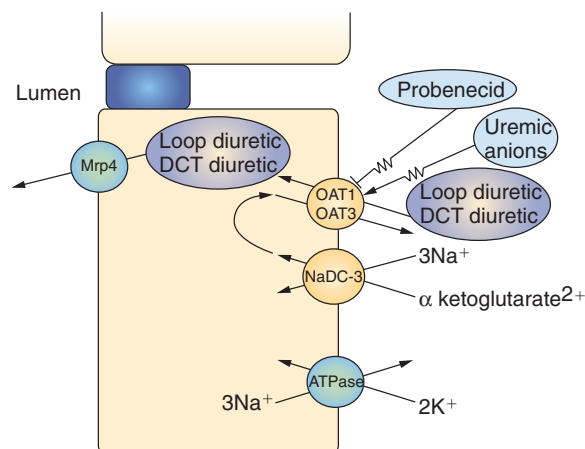


Fig. 50.5 Mechanisms of diuretic secretion by proximal tubule cells. Cell diagram of the S2 segment of the proximal tubule shows secretion of anionic diuretics, including loop diuretics and distal convoluted tubule (DCT) diuretics. Peritubular uptake by an organic anion transporter (primarily OAT1, although OAT3 may play a smaller role) occurs in exchange for α-ketoglutarate, which is brought into the cell by the Na⁺-dependent cation transporter NaDC-3. Luminal secretion can occur via a voltage-dependent pathway or in exchange for luminal hydroxyl or urate. A portion of the luminal transport traverses multidrug resistance-associated protein 4 (Mrp4). ATPase, Adenosine triphosphatase.

Once absorbed, loop diuretics circulate largely bound to albumin (91%–99%), greatly limiting their clearance by glomerular filtration. The diuretic volume of distribution varies inversely with the serum albumin concentration,¹⁰⁵ but this is not usually a major determinant of diuretic responsiveness (see later).¹⁰⁶ The metabolism of loop diuretics is comprised of both hepatic and renal mechanisms; the relative fractions that are cleared by each mechanism differ among agents. Loop diuretics, thiazides, and CAIs are all secreted avidly by a probenecid-sensitive organic anion transporter in proximal tubule cells (Fig. 50.5).^{107,108} Diuretics gain access to tubular fluid almost exclusively by proximal secretion. Studies have characterized this weak organic anion (OA[−]) transport process. Four isoforms of an OA transporter (OAT) have been cloned and are expressed in the kidney.^{108,109} Peritubular uptake by an OAT is a tertiary active process (see Fig. 50.5). Energy derives from the basolateral Na⁺-K⁺-ATPase providing a low intracellular [Na⁺] that drives an uptake of Na⁺ coupled to α-ketoglutarate (α-KG) to maintain a high intracellular level of α-KG. This in turn drives a basolateral OA-α-KG countertransporter. OAT1 is expressed on the basolateral membrane of the S2 segment of the proximal tubule.¹¹⁰ A mouse colony deficient in OAT1 was generated and shown to exhibit dramatically impaired renal OA secretion and furosemide resistance.¹¹¹ A similar effect was observed in OAT3-deficient mice, suggesting that both OAT1 and OAT3 mediate the secretion of loop diuretic by proximal cells, and that a lack of either is not fully compensated by the other.¹¹²

OATs translocate diuretics into the proximal tubule cell, where they can be sequestered in intracellular vesicles. They are secreted across the luminal membrane by a voltage-driven OA transporter¹¹³ and by a countertransporter in exchange for urate or OH[−].¹⁰⁹ The orphan transporter hNPT4 (human sodium phosphate transporter 4; SLC17A3) has been

identified as an organic anion efflux transporter that likely also secretes furosemide and bumetanide.¹¹⁴ In addition, the multidrug resistance-associated protein 4 (MRP4) has been identified as the third type of transporter involved in the urinary excretion of diuretics. Mice lacking MRP4 exhibited an almost twofold lower excretion of furosemide and hydrochlorothiazide.¹¹⁵ Approximately 50% of furosemide is eliminated by metabolism to the inactive glucuronide. Only the unmetabolized and secreted fraction is available to inhibit NaCl reabsorption. In contrast, bumetanide and torsemide are metabolized in the liver.^{116,117} Slow-release furosemide is more effective in reducing blood pressure and treating edema, highlighting the importance of pharmacokinetics in diuretic responsiveness.¹¹⁸ Torsemide's action is approximately twice as long as furosemide's,¹¹⁹ and bumetanide's action is shorter than that of furosemide. These differences may be clinically relevant.^{98,120} Unlike the elimination of bumetanide or torsemide,¹¹⁹ the elimination of furosemide in patients with CKD is greatly reduced because its metabolism to the inactive glucuronide occurs in the kidney; in contrast, metabolic inactivation of bumetanide and torsemide occurs mainly in the liver and therefore they are unaffected by uremia.⁴ This difference prolongs the $t_{1/2}$ of furosemide in CKD, leading to drug accumulation. However, the fraction of a dose excreted unchanged in patients with CKD is greater for furosemide, leading to an enhanced natriuretic response (Fig. 50.6). There is therefore a tradeoff in the selection of a loop diuretic in CKD: Furosemide can accumulate and cause ototoxicity at high doses, whereas bumetanide retains its metabolic inactivation but is therefore somewhat less potent.

Renal clearance of the active form of loop diuretics is reduced in CKD in proportion to the creatinine clearance.¹²¹ There is competition both for peritubular uptake¹⁰⁸ and for luminal secretion¹¹³ with other OAs, including urate, which accumulate in uremia. Metabolic acidosis depolarizes the membrane potential of proximal tubule cells,¹²² which decreases OA secretion,¹¹³ an effect that may explain why diuretic secretion is enhanced by alkalosis.¹²³ Therefore, the

increased plasma levels of OAs and urate and the metabolic acidosis of CKD impair proximal tubule secretion of diuretics and, hence, impair their delivery to their active sites in the nephron.

Proximal secretion of active furosemide is potentiated by albumin.¹²⁴ In the rabbit, an equal fraction of administered furosemide is taken up by probenecid-sensitive mechanisms in the S2 (secretory) or S1 segment of the proximal tubule, where it is conjugated and excreted as the inactive glucuronide (Fig. 50.7).¹²⁵ Unlike the uptake and secretion of active furosemide by the S2 segment, uptake and metabolism by the S1 segment is enhanced by a drop in albumin concentration. Therefore, a low serum albumin concentration enhances furosemide metabolism¹²⁶ yet decreases tubular secretion of active diuretic.¹²⁴ The consequences of this process are described later (see "Nephrotic Syndrome").

There is a similar sigmoidal relation between fractional sodium excretion and the log of the urinary diuretic concentration (Fig. 50.8). Inhibition of proximal secretion with probenecid shifts the curve of the plasma dose-response to the right but does not perturb the relationship between natriuresis and diuretic excretion.¹²⁷ Thus, natriuresis is related to the urinary concentration, but not the plasma concentration, of diuretic. The administration of indomethacin or other nonsteroidal antiinflammatory drugs (NSAIDs) reduces the responsiveness of the tubule to furosemide.¹²⁸ This reduction is due predominantly to reduced generation of PGE₂, because a natriuretic response to furosemide can be restored in indomethacin-treated rats by the infusion of PGE₂.¹²⁹ Both a reduced dietary salt intake and repeated administration of furosemide during salt restriction¹⁰⁴ diminish the renal tubular response to furosemide (see Fig. 50.8).

Although knowledge of the pharmacogenetics of diuretics is still rudimentary, a number of studies have demonstrated that certain polymorphisms contribute to individual differences in the response to loop diuretics. For example, in 97 healthy whites, one study reported that the acute effects of loop diuretics were greater in subjects with polymorphisms

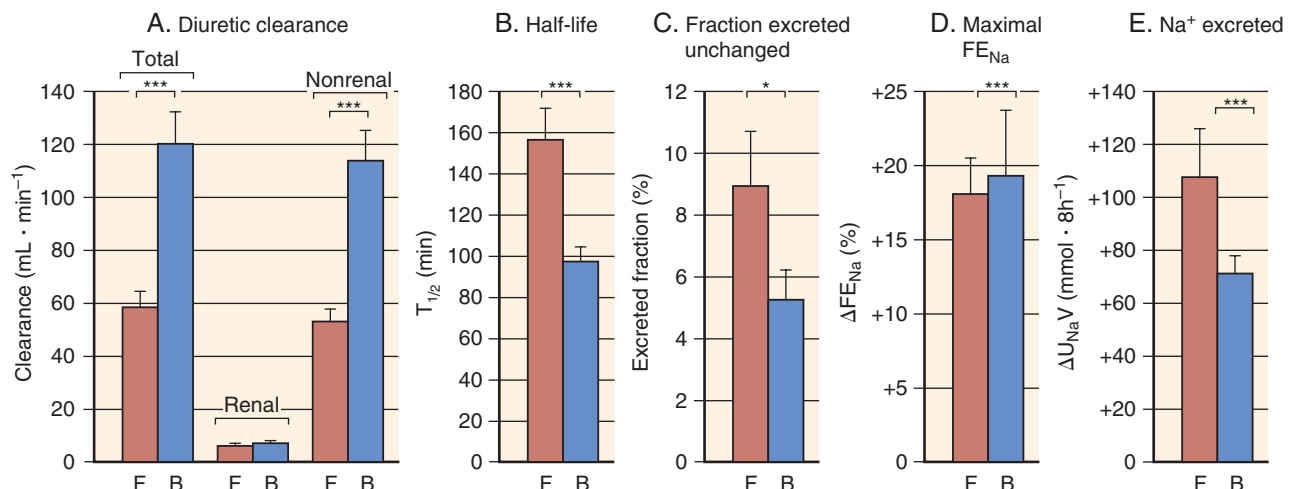


Fig. 50.6 (A–E) Comparison of the pharmacokinetics and dynamics of furosemide (F, 160 mg; metabolically inactivated in the kidney) and bumetanide (B, 4 mg; metabolically inactivated in the liver) in 10 subjects with chronic kidney disease (mean creatinine clearance, 12 ± 2 mL/min). Significance of difference: *, $P < .05$; ***, $P < .005$. FE_{Na} , Fractional excretion of sodium; $T_{1/2}$, half-life; U_{Na} , urinary sodium. (Redrawn from data in Voelker JR, Cartwright-Brown D, Anderson S, et al. Comparison of loop diuretics in patients with chronic renal insufficiency. *Kidney Int.* 1987;32:572–578.)

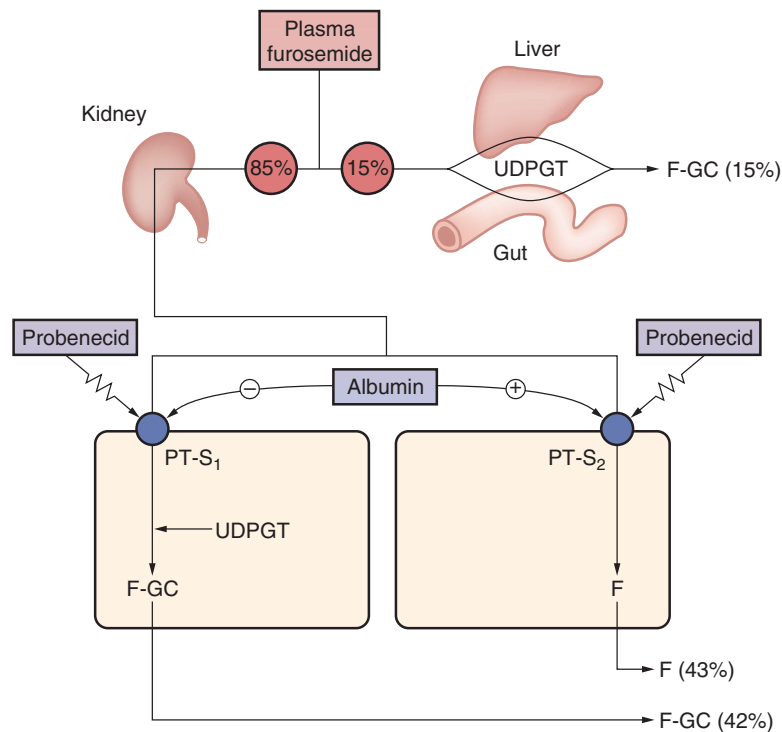


Fig. 50.7 Diagrammatic representation of the disposition of intravenous furosemide and the effects of hypoalbuminemia or probenecid in normal or hypoalbuminemic rabbits. After intravenous administration of furosemide, 15% is metabolized by uridine diphosphate glucuronyl transferase (*UDPGT*) in the liver and gut to the inactive furosemide glucuronide (*F-GC*). Of the remainder, 85% is transported by the kidney. Some 42% is taken up in the S1 segment of the proximal tubule (*PT-S₁*) and metabolized to the inactive glucuronide, and the remainder is taken up by the S2 segment (*PT-S₂*) and secreted in active form into the lumen. Both uptake processes are inhibited by probenecid. The plasma albumin concentration facilitates uptake and secretion by *PT-S₂* but inhibits uptake and metabolism by *PT-S₁*. (Modified from Pichette V, Geadah D, du Souich P. The influence of moderate hypoalbuminemia on the renal metabolism and dynamics of furosemide in the rabbit. *Br J Pharmacol*. 1996;119:885–890.)

RENAL RESPONSE TO FUROSEMIDE IN HUMANS: EFFECTS OF INDOMETHACIN, SALT RESTRICTION, AND REPEATED FUROSEMIDE ADMINISTRATION

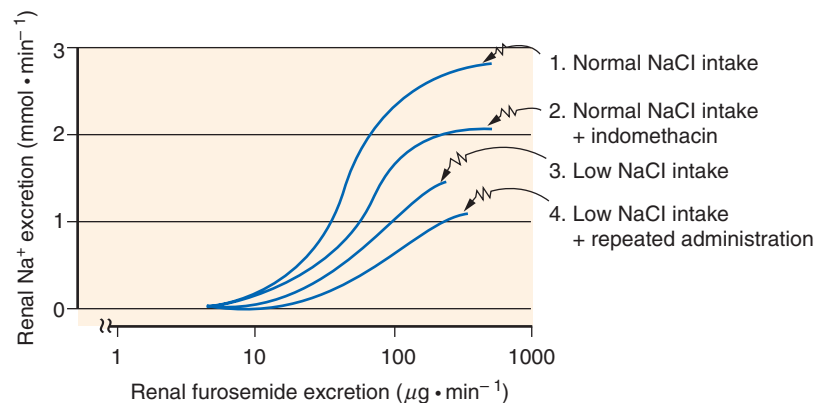


Fig. 50.8 Relationship between the excretion of Na⁺ and furosemide (log scale) following a bolus intravenous injection of 40 mg furosemide in normal subjects with a normal NaCl intake (1), with a normal NaCl intake after indomethacin (2), with a low Na⁺ intake (20 mmol/24 hours) (3), and for the third day of furosemide administration with a low Na⁺ intake (4). (Modified from Wilcox CS, Mitch WE, Kelly RA, et al. Response of the kidney to furosemide. *J Lab Clin Med*. 1983;102:450–458; and Chennavasin P, Seiwel R, Brater DC. Pharmacokinetic-dynamic analysis of the indomethacin-furosemide interaction in man. *J Pharmacol Exp Ther*. 1980;215:77–81.)

in the genes encoding NCC and the beta subunit of the epithelial Na⁺ channel (ENaC), but smaller in those with a polymorphism in the gene encoding the gamma subunit of ENaC.¹³⁰ This finding suggests that individual variations in loop diuretic response may, in part, be attributed to lower

or higher activities of transporters located distal to NKCC2. Another pharmacokinetic study has identified female gender and polymorphisms in the gene encoding the organic anion transporter OATP1B1 as predictors for the slower elimination of torsemide.¹³¹

CLINICAL INDICATIONS

Clinical indications for the use of loop diuretics are discussed later (see “[Clinical Uses of Diuretics](#)”).

ADVERSE EFFECTS

Adverse effects of loop diuretics are discussed later (see “[Adverse Effects of Diuretics](#)”).

THIAZIDES AND THIAZIDE-LIKE DIURETICS (DISTAL CONVOLUTED TUBULE DIURETICS)

SITES AND MECHANISMS OF ACTION

Thiazides and thiazide-like diuretics are moderately active drugs that increase the excretion of sodium, chloride, and potassium while reducing the excretion of calcium. The major site of action of thiazide and thiazide-like diuretics is the distal convoluted tubule (DCT), where they block coupled reabsorption of Na^+ and Cl^- (Fig. 50.9).^{73,96,132} The true thiazides (benzothiadiazines) are chlorothiazide, hydrochloro-

thiazide, bendroflumethiazide, and others. Subsequent to their development, nonthiazide drugs with similar activities were developed. Substitution of the ring sulfone in the thiazides with a carbonyl group provides a group of quinazolinones with diuretic activity similar to that of the thiazides (e.g., metolazone). Chlorthalidone, a substituted benzophenone that does not contain the benzothiadiazine molecular structure, exhibits strong CA activity and a prolonged half-life and has seen widespread use as an antihypertensive.

The predominant effect of the thiazide diuretics is to inhibit the thiazide-sensitive NCC (gene symbol *SLC12A3*). This protein is expressed in the DCT^{133,134} and is inhibited directly by thiazides (see later). Several thiazides and thiazide-like drugs (e.g., chlorothiazide, hydrochlorothiazide, chlorthalidone) also inhibit CA, contributing to their natriuretic efficacy.¹³⁵ However, patients with Gitelman syndrome, who have a loss-of-function mutation in the NCC, and mice lacking NCC demonstrate a dramatically impaired natriuretic response to thiazides,¹³⁶ confirming that the principal effect of these drugs is to inhibit NCC. A small component of thiazide-sensitive

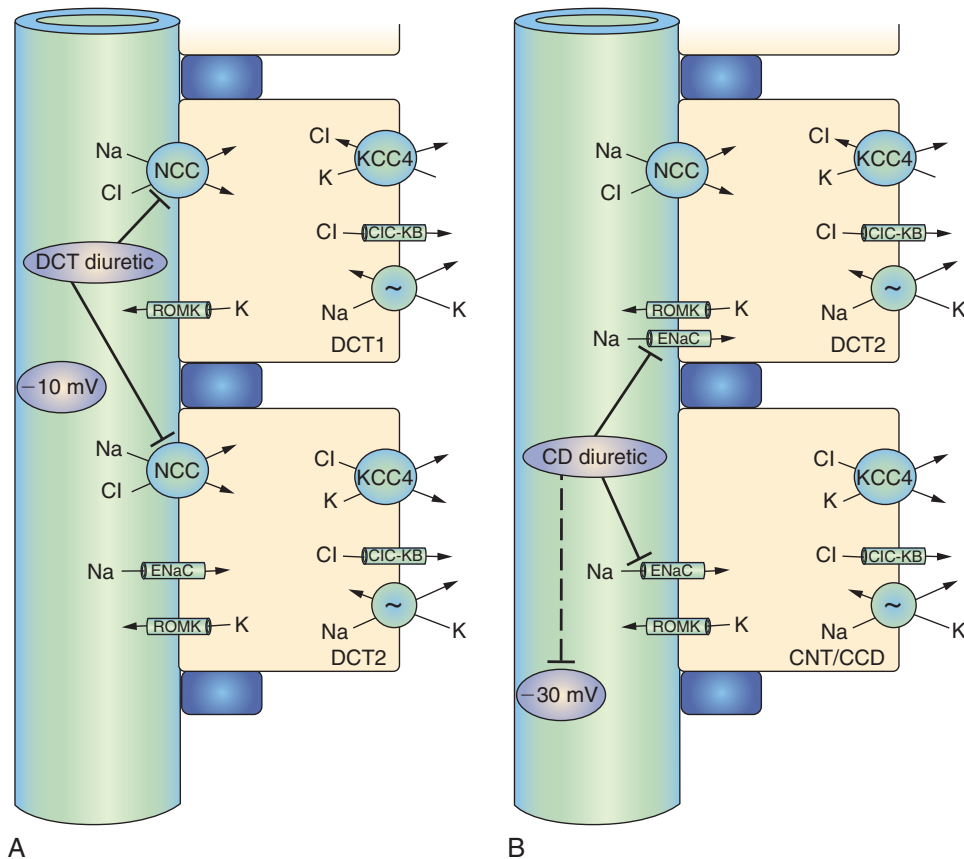


Fig. 50.9 Mechanisms of distal convoluted tubule (DCT) and collecting duct (CD) diuretics. (A) Mechanism of action of DCT diuretics. In rat, mouse, and human, two types of DCT cells have been identified, referred to here as DCT1 and DCT2. Na^+ and Cl^- are reabsorbed across the apical membrane of DCT1 cells only via the thiazide-sensitive $\text{Na}^+\text{-Cl}^-$ cotransporter (NCC). This transport protein is also expressed by DCT2 cells where Na^+ can also cross through the epithelial Na^+ channel (ENaC; see text for details). Thus, the transepithelial voltage along the DCT1 is near to 0 mV, whereas it is finite and lumen-negative along the DCT2. (B) Mechanism of action of CD diuretics. The late distal convoluted tubule cells (DCT2 cells) and connecting tubule (CNT) or cortical collecting duct (CCD) cells are shown. Na^+ is reabsorbed via ENaC, which lies in parallel with a renal outer medullary K^+ channel (ROMK). The transepithelial voltage is oriented with the lumen negative, relative to the interstitium (shown in the circled value), generating a favorable gradient for transepithelial K^+ secretion. Drugs that block the epithelial Na^+ value reduce the voltage toward 0 mV (effect indicated by dashed line), thereby inhibiting K^+ secretion. CIC-KB, Chloride channel protein; KCC4, Potassium-chloride cotransporter 4.

Na⁺ excretion was observed in mice lacking NCC, which has been attributed to the inhibition of a sodium-dependent chloride-bicarbonate exchange process.¹³⁷ Furthermore, Na⁺ reabsorption by the proximal tubule is enhanced during long-term treatment with thiazides, even when the drug has significant CA-inhibiting capacity.¹³⁸

Like loop diuretics, diuretics that are active in the DCT, including the thiazides, are organic anions that bind to the transport protein from the luminal surface. The mechanism(s) of NCC inhibition have been studied using two approaches. First, Beaumont and Tran and colleagues have shown that [³H]-metolazone binds avidly to kidney membrane proteins; its binding is inhibited competitively by Cl[−], suggesting that Cl[−] and diuretics compete for the same binding site.^{139,140} These results are reminiscent of studies that used [³H]-bumetanide to study properties of the NKCC proteins and were used to develop a kinetic model for the NCC.¹⁴¹ Moreno and colleagues have expressed chimeras of the NCC in *Xenopus* oocytes and defined thiazide affinity on the basis of transport inhibition. The results suggest a more complicated picture. They concluded that thiazide diuretic affinity is conferred by transmembrane segments 8 through 12, whereas transmembrane segments 1 through 7 affect chloride affinity. Both domains are involved in determining Na⁺ affinity.¹⁴² These data suggest that the affinity of thiazide diuretics for binding to the transport protein is in a region distinct from the region that participates in Cl[−] transport.

Thiazides increase potassium excretion, but do not augment DCT secretion of K⁺ directly.^{96,143} Instead, their effects result from their tendency to stimulate aldosterone secretion, increase distal flow, and increase calcium reabsorption.¹⁴⁴ Mineralocorticoids, glucocorticoids,¹⁴⁵ and estrogens¹⁴⁶ enhance thiazide binding and tubular actions. Thiazides reduce Ca²⁺ excretion. Three potential and nonredundant mechanisms have been postulated (see Fig. 50.4).^{147,148} First, blockade of luminal NaCl entry reduces the tubule cell intracellular [Na⁺] sufficiently to enhance basolateral Na⁺-Ca²⁺ exchange.¹⁴⁹ Second, thiazide-induced blockade of luminal NaCl entry reduces cell Cl[−] concentration, thereby hyperpolarizing the membrane voltage (making the interior of the cell more negative, electrically). Hyperpolarization increases calcium entry via the transient receptor potential channel subfamily V, member 5 (TRPV5) channel, which is expressed at the apical membrane of DCT and connecting tubule cells.^{150,151} Third, thiazides stimulate proximal reabsorption of Ca²⁺ owing to ECV depletion.¹⁵² The importance of this proximal effect has been highlighted because thiazides reduce Ca²⁺ excretion, even when TRPV5, the major distal calcium reabsorptive pathway, has been knocked out and is absent.¹⁵³

However, other mechanisms in the DCT must also play a role because a study in mice deficient for the DCT-specific protein parvalbumin has shown that hydrochlorothiazide does not increase natriuresis but does induce hypocalciuria.¹⁵⁴ Similarly, a study in humans has shown that hypovolemia is not the sole cause of hypocalciuria in patients with Gitelman syndrome who have a genetic inactivation of NCC.¹⁵⁵ Thiazides produce a sustained reduction in renal Ca²⁺ excretion that is accompanied by a small rise in serum Ca²⁺ concentration. Mg²⁺ excretion is enhanced by thiazide diuretics, at least during prolonged therapy¹⁵⁶ (see Fig. 50.4). It has also been shown that transient receptor potential channel melastatin 6 (TRPM6) is a magnesium channel of the distal nephron.^{157,158}

Long-term thiazide treatment of mice diminishes TRPM6 mRNA expression modestly and reduces TRPM6 protein abundance by approximately 80%. Such changes would be expected to reduce magnesium reabsorption along the distal nephron, leading to Mg²⁺ wasting. Mg²⁺ depletion that can occur during chronic thiazide administration may be augmented by K⁺ depletion.¹⁵⁶ Thiazides reduce urate clearance secondary to ECV depletion⁸² and competition for tubular uptake.¹⁰⁷

Expression of the water channel aquaporin-2 (AQP2) begins at the junction between the DCT and connecting tubule. Thus, the NCC-expressing DCT comprises the terminal diluting segment of the kidney. Thiazides impair maximal urinary dilution but not maximal urinary concentration.¹⁵⁹ Thiazides also enhance water absorption from inner medullary collecting ducts in an AVP-independent manner.¹⁶⁰ This effect is correlated with an increase in AQP2 expression during long-term thiazide treatment.¹⁶¹ These effects may contribute to the tendency of thiazide diuretics to produce hyponatremia. Central effects on thirst, however, may also contribute (see later discussion of adverse effects).

Clinical Relevance

Implications of Differential Effect of Diuretics on Electrolyte and Acid-Base Handling

The classes of diuretics have different effects on acid-base and electrolyte balance. Hyponatremia is a specific adverse effect of thiazide diuretics. Loop and thiazide diuretics increase urinary magnesium and potassium excretion, whereas distal K⁺-sparing diuretics reduce this. Urinary calcium excretion is increased by loop diuretics and decreased by thiazide diuretics. All diuretics tend to cause metabolic alkalosis, except for carbonic anhydrase inhibitors, which cause metabolic acidosis. Diuretic combinations may be used to prevent clinically significant acid-base and electrolyte disturbances.

PHARMACOKINETICS OF AND DIFFERENCES AMONG THIAZIDES

Thiazides are readily absorbed and are extensively bound to plasma proteins. They are eliminated largely through secretion by the S2 segment of the proximal tubule, mostly via OAT1 and OAT3.^{108,112} The t_{1/2} is prolonged in renal failure and in older adults, reducing natriuretic efficacy.¹⁶² The more lipid-soluble drugs (e.g., bendroflumethiazide, polythiazide) are more potent, have a more prolonged action, and are more extensively metabolized.¹⁶³ Chlorthalidone has a particularly prolonged action.¹⁶³ Indapamide is sufficiently metabolized to limit accumulation in renal failure.¹⁶³ Extrarenal effects of thiazide diuretics, including effects on platelet aggregation and vascular permeability, vary among the types of thiazide diuretics. This may explain why thiazide-like diuretics (e.g., chlorthalidone, indapamide) appear to have superior cardioprotective effects compared with thiazide-type diuretics (e.g., chlorothiazide, hydrochlorothiazide).^{164–166}

As stated above, little is known about the pharmacogenetics of diuretics, but available studies have suggested that differences in individual responses are important. In one study, diuretic therapy in carriers of a variant of α -adducin, a

cytoskeleton protein important for the function of renal $\text{Na}^+\text{-K}^+\text{-ATPase}$, was found to be associated with a lower risk of combined myocardial infarction and stroke than other antihypertensive therapies.¹⁶⁷ In another study, polymorphisms in with-no-lysine kinase 1 (WNK1), a kinase involved in the regulation of NCC, also affected the response to a thiazide diuretic.¹⁶⁸ Finally, polymorphisms in Nedd4-2, a ubiquitin ligase that regulates the sodium chloride cotransporter,¹⁶⁹ predicts the blood pressure response to hydrochlorothiazide in white subjects and cardiovascular outcome with hydrochlorothiazide in black subjects.¹⁷⁰

CLINICAL INDICATIONS AND ADVERSE EFFECTS

The clinical indications for the use of thiazides and thiazide-like drugs are discussed below (see “[Clinical Uses of Diuretics](#)”). Adverse effects of thiazides and thiazide-like drugs are discussed below (see “[Adverse Effects of Diuretics](#)”).

DISTAL POTASSIUM-SPARING DIURETIC AGENTS

Distal K^+ -sparing diuretics comprise those that directly block the ENaC (amiloride and triamterene) and antagonists of the mineralocorticoid receptor (spironolactone and eplerenone). Finerenone is the first of a class of dihydropyridine derivatives with antimineralocorticoid actions currently under development.¹⁷¹

SITES AND MECHANISMS OF ACTION

Distal K^+ -sparing diuretics act on the cells in the late DCT, connecting tubule, and cortical CD (the ASDN), where they inhibit luminal Na^+ entry via ENaC (see [Fig. 50.9](#)).^{132,172} They depolarize the lumen-negative transepithelial voltage, diminishing the electrochemical gradient for K^+ and H^+ secretion.^{132,173}

Both amiloride and triamterene are organic cations that block ENaC directly from the luminal surface. Amiloride also inhibits NHE3, but the affinity of amiloride for NHE3 is low enough that the distal effects predominate in clinical use. In experimental work, congeners of amiloride that are more selective for ENaC or NHE3 have been developed, although they have not been used clinically. Amiloride appears to bind ENaC in its conducting pore and is thus a pore blocker.¹⁷⁴ Amiloride binding is sensitive to the electric field, and the agent appears to compete with Na^+ for binding to the pore of the channel.¹⁷⁵ Amiloride may interact with several regions on the ENaC protein, but one amiloride-binding region comprises a short amino acid stretch within the extracellular loop.¹⁷⁶ In addition to their natriuretic effects, amiloride and triamterene also reduce the excretion of Ca^{2+} and Mg^{2+} .^{156,177}

Spironolactone and eplerenone are competitive antagonists of the mineralocorticoid receptor. Mineralocorticoid receptor antagonists (MRAs; also called “aldosterone receptor antagonists”) were developed when it was discovered that aldosterone is an 18-aldehyde derivative of corticosterone and that progesterone increases Na^+ excretion by blocking exogenously administered mineralocorticoid. Eventually, spironolactone was developed. It was found not to have any effect on urinary Na^+ or K^+ directly, but instead competitively blocks the mineralocorticoid receptor. Structurally, spironolactone strongly resembles aldosterone. Eplerenone was developed as an attempt to find an MRA with fewer estrogenic side effects.

These drugs were used for many years primarily to reduce the excretion of K^+ and net acid, especially when used in combination with other diuretics, because these drugs cause a very modest natriuresis. Under certain circumstances, however, their natriuretic efficacy can be significant. For example, spironolactone is more effective than furosemide in reducing cirrhotic ascites.¹⁷⁹ Furthermore, spironolactone is often an effective adjunct in the treatment of resistant hypertension.¹⁸⁰ This agent has achieved an important role in the treatment of heart failure with a reduced ejection fraction,¹⁸¹ although the mechanisms whereby it achieves protection continue to be debated.

PHARMACOKINETICS

Triamterene is well absorbed. It is rapidly hydroxylated to active metabolites.¹⁸² The drug and its metabolites are secreted by the organic cation pathway in the proximal tubule,³ with half-lives of 3 to 5 hours. Triamterene and its active metabolites accumulate in patients with cirrhosis because of decreased biliary secretion,¹⁸³ and in older adults^{184,185} and patients with CKD¹⁶² because of decreased renal excretion.

Amiloride is incompletely absorbed. Its duration of action is approximately 18 hours. It is secreted into the tubular fluid by the organic cation transport pathway.¹⁸⁶ Other organic cations, such as cimetidine, inhibit its secretion and prolong its half-life.¹⁸⁶ It accumulates in renal failure¹⁸⁷ and may worsen renal function.¹⁸⁸ Spironolactone is readily absorbed and circulates bound to plasma proteins. Its intrinsic half-life is short, but it is metabolized to active compounds with considerably prolonged actions. Spironolactone is metabolized to canrenones ($t_{1/2} = 16$ hours) and to sulfur-containing metabolites, predominantly 7- α -thiomethylspironolactone ($t_{1/2} = 13$ hours).¹⁸⁹ Canrenones are metabolized by the cytochrome P-4503A system.¹⁹⁰ Clinically, spironolactone has a $t_{1/2}$ of approximately 20 hours. It takes 10 to 48 hours to become maximally effective.¹⁹¹ It is lipid-soluble and enters distal renal tubules from the plasma.¹⁹² Eplerenone has fewer antiandrogenic and proestrogenic effects.^{193,194} However, it is metabolized with a half-life of 3 hours and therefore should be given twice daily.¹⁹⁵

CLINICAL INDICATIONS

Distal K^+ -sparing agents are used to prevent or treat hypokalemic alkalosis, especially in combination with a thiazide diuretic.¹⁹⁶ Amiloride can prevent amphotericin-induced hypokalemia and hypomagnesemia.¹⁹⁷ Spironolactone is indicated as a first-line agent for ECV expansion in the setting of cirrhotic ascites, in which it is more effective than daily furosemide,^{198–200} although current guidelines suggest that it should be used in combination with furosemide at a dose ratio of spironolactone 100 mg/furosemide 40 mg.²⁰¹ It is also indicated for heart failure with a reduced ejection fraction, in which its effects may include renal and extrarenal mineralocorticoid receptor blockade,²⁰² and recommended doses are limited to 25 and 50 mg/day, respectively. Spironolactone is also commonly used to treat hypertension associated with hyperaldosteronism and for resistant hypertension,^{180,203,204} and it may reduce proteinuria and progressive loss of kidney function in CKD.^{205,206} However, see the following discussion of adverse effects. Eplerenone is indicated to prevent cardiac remodeling and systolic dysfunction in the

setting of recent myocardial infarction.^{207,208} Preliminary experimental data have suggested that spironolactone may help prevent the development of CKD after ischemic AKI (AKI-induced CKD).²⁰⁹

ADVERSE EFFECTS AND DRUG INTERACTIONS

Hyperkalemia is the most common complication of the distal K^+ -sparing diuretics. The risk is dose-dependent and increases considerably in patients with CKD or in those receiving K^+ supplements, angiotensin-converting enzyme (ACE) inhibitors, angiotensin receptor blockers (ARBs), NSAIDs, beta-blockers, heparin, or ketoconazole. The incidence of hyperkalemia-associated morbidity and mortality in Canada rose sharply after the publication of the Randomized Aldactone Evaluation Study (RALES). This study demonstrated the efficacy of spironolactone in improving outcomes in heart failure and may relate to the consequent widespread use of this agent in patients with congestive heart failure and impaired renal function.²¹¹ It is important to note that the original RALES study specifically excluded patients with several of these comorbidities. Renal failure appears to be another complication in this group.²¹²

Gynecomastia may occur in men, especially as the dose is increased²¹³ but, even at low doses,²¹⁴ decreased libido and impotence have also been reported. Women may experience menstrual irregularities, hirsutism, or swelling and tenderness of the breast. Impaired net acid excretion can cause metabolic acidosis,²¹⁵ which worsens hyperkalemia.

Amiloride and triamterene accumulate in renal failure,^{185,216} and triamterene accumulates in cirrhosis.¹⁸³ Therefore, these drugs should be avoided in patients with these conditions. Triamterene occasionally precipitates in the urinary collecting system and causes obstruction.²¹⁷ It can cause acute kidney injury when given with indomethacin.²¹⁸

MISCELLANEOUS AGENTS

DOPAMINERGIC AGENTS

When given to normal subjects in low doses (1–3 $\mu\text{g}/\text{kg}$ per minute), dopamine causes a modest increase in the GFR, reduces proximal reabsorption via a cAMP-induced inhibition of the $\text{Na}^+\text{-H}^+$ antiporter, and increases Na^+ excretion.²¹⁹ Fenoldopam is a selective dopamine type 1 receptor agonist, with little cardiac stimulation.²¹⁹ Unfortunately, these beneficial effects are reduced in patients who are critically ill and/or receiving vasopressors.²²⁰ A comprehensive review of the literature has concluded that in controlled trials, low-dose dopamine universally fails to improve renal outcomes in patients at high risk for AKI and that in the largest trials, it had no effect on renal function, need for dialysis, or mortality in critically ill patients with early renal dysfunction.²²⁰ Thus, there is currently no justification for the use of low-dose dopamine for renal protection.²²¹ Dopamine infusion at higher rates has a role as a pressor agent in septic shock or refractory heart failure, but the benefits can be offset by arrhythmias.²²⁰

VASOPRESSIN RECEPTOR ANTAGONISTS

Vasopressin receptor antagonists are nonpeptide molecules that competitively inhibit one or more of the human vasopressin receptors, V1aR, V1bR, and V2R.²²² Conivaptan is a combined V1aR/V2R antagonist for intravenous use,

whereas tolvaptan, mozavaptan, and lixivaptan are orally active V2R-selective antagonists. All these agents cause a free water diuresis without appreciable natriuresis or kaliuresis and they are therefore sometimes referred to as “aquaretics.”²²³ This effect is mainly attributed to the inhibition of V2R in the collecting duct, which prevents vasopressin from recruiting AQP2 water channels to increase water reabsorption. Therefore, vasopressin receptor antagonists can be used to treat hypervolemic or euvolemic hyponatremia, in which increased vasopressin is considered inappropriate. Co-inhibition of V1aR, which is located in vascular smooth muscle, could be beneficial to reduce coronary vasoconstriction, myocyte hypertrophy, and vascular resistance in patients with heart failure,²²⁴ but definitive studies on this effect are lacking. At present, some 20 clinical trials have tested these agents against placebo or conventional therapy in patients with liver cirrhosis, heart failure, or hyponatremia secondary to SIADH.²²⁵ In all trials, vasopressin receptor antagonists effectively raised serum sodium levels and helped correct hyponatremia. In addition, a positive effect on some secondary end points was observed in patients with heart failure, including improved mental condition and reductions in body weight, dyspnea, and ascites.^{226–228} However, the Efficacy of Vasopressin Antagonist in Heart Failure Outcome Study with Tolvaptan (EVEREST), which involved 4133 patients hospitalized for heart failure (with or without hyponatremia), did not show a beneficial effect of tolvaptan on the primary outcome of death and rehospitalization for heart failure.²²⁷ Thus, vasopressin receptor antagonists appear effective in the correction of hyponatremia but have not yet shown an effect on primary outcomes.

ADENOSINE TYPE 1 RECEPTOR ANTAGONISTS

Aminophylline is an adenosine receptor antagonist that inhibits NaCl reabsorption in the proximal tubule and diluting segments and causes a modest increase in GFR.²²⁹ Highly selective adenosine 1 (A1) receptor antagonists are natriuretic²³⁰ and antihypertensive, and they potentiate furosemide-induced natriuresis in normal humans²³¹ and in patients with diuretic-resistant heart failure. A1 antagonists disrupt glomerulotubular balance and TGF, thereby decreasing proximal reabsorption and increasing the GFR.²³⁰ Several A1 receptor antagonists have been tested in patients with acute heart failure, but a large trial with the drug rolofylline did not show improvement in survival, heart failure status, or kidney function.²³²

NESIRITIDE

Nesiritide is the recombinant form of B-type natriuretic peptide, which can be administered intravenously in acute decompensated congestive heart failure (see later discussion of clinical uses of diuretics in congestive heart failure). By stimulating cGMP, this agent causes both a natriuresis and relaxation of smooth muscle.

NEPRILYSIN INHIBITORS

Neprilysin inhibitors prevent the breakdown of natriuretic peptides, and therefore contribute to natriuresis. The combined use of an ARB and neprilysin inhibitor (LCZ696) has been shown to reduce rates of cardiovascular death and hospitalization for heart failure in patients with class II, III, or IV heart failure.²³³

SODIUM GLUCOSE-LINKED TRANSPORTER 2 (SGLT2) INHIBITORS

Sodium glucose-linked transporter 2 (SGLT2) inhibitors block glucose reabsorption in the proximal tubule, thereby delivering much of the filtered glucose to the urine and inducing an osmotic diuresis. Empagliflozin, canagliflozin, and dapagliflozin are three SGLT2 inhibitors that are licensed for the treatment of type 2 diabetes mellitus. Because SGLT2 cotransports Na^+ with glucose, and the filtered load of glucose is increased in uncontrolled diabetes mellitus, a significant fraction of proximal tubular Na^+ reabsorption is blocked by these drugs, thereby delivering more Na^+ to the loop of Henle and macula densa segments.²³⁴ The result is increased renal Na^+ excretion and activation of TGF that increases preglomerular vascular tone and reduces the GFR. The resulting reduction in glomerular capillary pressure and glomerular blood flow reduces hyperfiltration and, in combination with a fall in blood pressure, may account for the slowing or prevention of the progressive loss of the GFR in patients with diabetic nephropathy treated with empagliflozin.²³⁵ Moreover, the prolonged increase in Na^+ delivery from the proximal tubule leads to an adaptive increase in NaCl reabsorption in the loop of Henle²³⁴ that could account for the observation of an adaptive synergistic natriuresis when a loop diuretic is given to subjects adapted to an SGLT2 inhibitor.²³⁶ Thus, the addition of an SGLT2 inhibitor may help relieve loop diuretic resistance in patients with diabetes mellitus while providing some protection from progressive loss of the GFR.^{237,238}

NOVEL DIURETICS IN PRECLINICAL DEVELOPMENT

Several novel diuretics are being developed targeting urea channels, ROMK, or pendrin in the kidney. Urea channel inhibitors induce an osmotic diuresis by inhibiting urea reabsorption in the renal medulla.²³⁹ An inhibitor of the urea channel UT-B (SLC14A1) increased urine output in rats without commensurate urinary sodium and potassium loss.²⁴⁰ Recently, a small-molecule screen also identified selective inhibitors of UT-A1 (SLC14A2).²⁴¹

Dimethylthiourea was found to inhibit both UT-B and UT-A1 and produced a greater diuresis in rats than furosemide while preserving sodium levels better. This compound was also effective in preventing hyponatremia in a model of SIADH.²⁴² ROMK inhibitors induce a robust natriuresis in rats but, unlike loop and thiazide diuretics, do so without kaliuresis.²⁴³ A ROMK inhibitor caused greater diuresis and natriuresis when combined with hydrochlorothiazide or amiloride, but not bumetanide, suggesting that the diuretic target site of ROMK inhibitors is the TAL.²⁴³ Pendrin is a $\text{Cl}^-/\text{HCO}_3^-$ exchanger in intercalated cells involved in Na^+ reabsorption and acid-base balance.²⁴⁴ Small-molecule pendrin inhibitors have been identified from a high-throughput screen.²⁴⁵ Pendrin inhibitors alone are not effective diuretics.²⁴⁶ In contrast, the addition of a pendrin inhibitor to mice treated with furosemide further increased urine output by 30% to 60%.²⁴⁶ This confirms the importance of pendrin upregulation in chronic treatment with diuretics that target Na^+ transport upstream.²⁴⁷ Accordingly, pendrin inhibitors may prove useful compounds for combined diuretic treatment or in diuretic resistance (see below).

ADAPTATION TO DIURETIC THERAPY

Diuretics entrain a set of homeostatic mechanisms that limit their fluid-depleting actions and contribute to resistance to these agents and their adverse effects.

DIURETIC BRAKING PHENOMENON

The first dose of a diuretic normally produces a reassuring diuresis. However, in normal subjects, a new equilibrium is rapidly attained when body weight stabilizes and daily fluid and electrolyte excretion no longer exceeds intake.¹⁰⁴ This reaction is called the “diuretic braking phenomenon.” The effects of dietary salt intake on the diuretic braking phenomenon during 3 days of loop diuretic administration to normal human subjects are shown in Fig. 50.10.^{104,248–250} During high Na^+ intake (270 mmol/24 hours), the first dose of furosemide (F_1) causes a large negative Na^+ balance over the ensuing 6 hours (blue bars in Fig. 50.10A), followed by 18 hours during which Na^+ excretion is reduced well below intake (postdiuresis salt retention), which results in a positive Na^+ balance (light green areas in Fig. 50.10A) that offsets the preceding negative Na^+ balance. The natriuresis caused by the third daily dose of furosemide (F_3) is comparable to that caused by the first dose and also is followed by a restoration of Na^+ balance. Consequently, at high levels of Na^+ intake, subjects regain neutral Na^+ balance within 24 hours of each dose of furosemide and maintain their original body weight. A similar diuretic braking phenomenon occurs during established furosemide therapy.⁸³ During severe dietary Na^+ restriction (20 mmol/24 hours; Fig. 50.10C), the first dose of furosemide produces a blunted natriuresis. However, Na^+ balance cannot be restored because of the low level of dietary Na^+ intake. Consequently, virtually all the Na^+ lost during the diuretic phase is represented as a negative Na^+ balance for the day. Unlike in the high-salt protocol, tolerance manifests as a 40% reduction in the natriuretic response to the drug over 3 days. However, despite a blunted initial response and the development of tolerance, all subjects lose Na^+ and body weight. A loop diuretic given during an Na^+ intake of 120 mmol/24 hours (equivalent to a salt-restricted diet) causes Na^+ loss, but the loss is curtailed by a combination of postdiuretic renal salt retention and diuretic tolerance (see Fig. 50.10B).²⁵⁰

Clinical Relevance

Diuretic Braking Phenomenon

The diuretic braking phenomenon refers to the observation that diuretics no longer produce a negative sodium balance. It is explained by postdiuretic renal salt retention and compensatory distal reabsorption. Dietary sodium restriction or the addition of a second diuretic are effective strategies to overcome the diuretic braking phenomenon.

Furosemide kinetics and the GFR are unchanged over 3 days of furosemide administration. During a low NaCl intake, the curve representing the relationship between natriuresis and furosemide excretion on a graph is shifted to the right

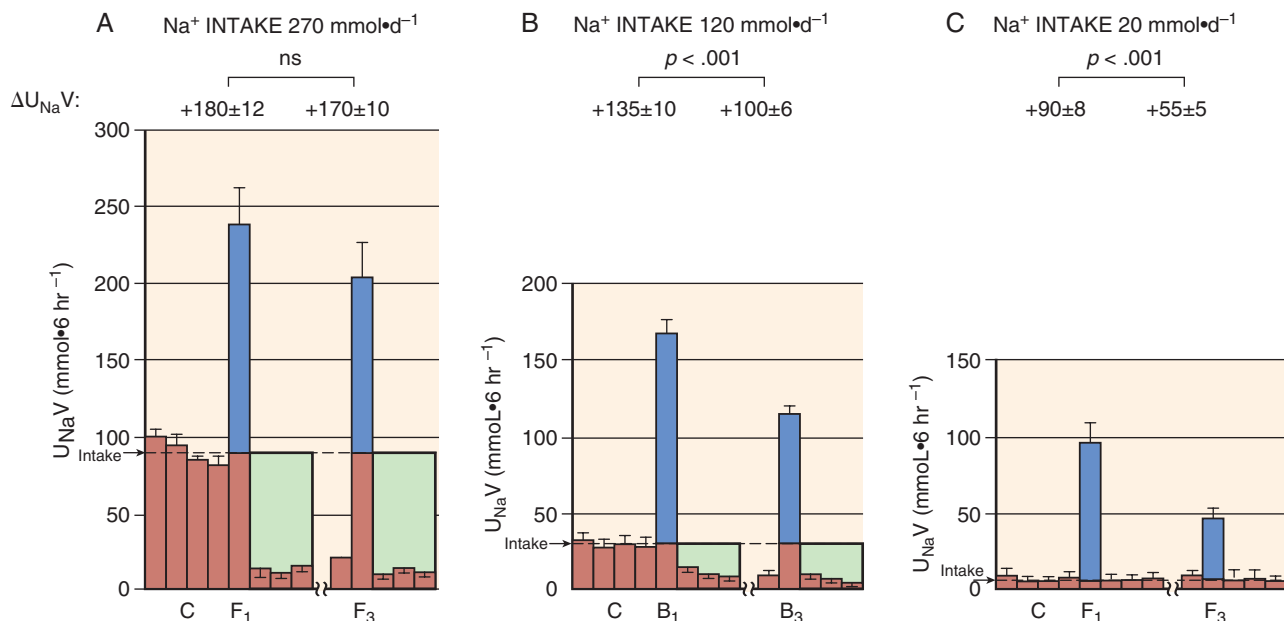


Fig. 50.10 Effects of dietary salt intake on the diuretic braking phenomenon. Renal Na⁺ excretion (mmol/6 hours) for 24 hours before and after the first (F₁) and third (F₃) daily doses of furosemide (A and C; 40 mg intravenously) and (B) bumetanide (B₁, B₃, 1 mg intravenously) in groups of 8 to 10 normal subjects equilibrated to fixed daily Na⁺ intakes. The average level of Na⁺ intake (mmol/6 hours) is shown by broken horizontal lines. Negative Na⁺ balance is indicated by blue bars and positive Na⁺ balance by light green areas. The mean $\pm$ standard error of the mean (SEM) values for diuretic-induced increases in Na⁺ excretion above baseline values ($\Delta U_{Na}V$) for 6 hours after the administration of the diuretic are shown at the top. (Modified from Wilcox CS, Mitch WE, Kelly RA, et al. Response of the kidney to furosemide. *J Lab Clin Med.* 1983;102:450–458.)

by the third day of diuretic administration (see Fig. 50.8), indicating a blunting of diuretic responsiveness.

One month of furosemide therapy for hypertension reduces the natriuretic response to a test dose of furosemide by 18%.⁸³ This tolerance cannot be ascribed to aldosterone nor to a fall in plasma or ECV, because tolerance to furosemide is not prevented by spironolactone and does not develop during thiazide therapy, which causes similar reductions in body fluids. In fact, the natriuretic response to a test dose of a thiazide is augmented during furosemide therapy. Thus, tolerance to furosemide is class-specific and depends on increased NaCl reabsorption at a downstream, thiazide-sensitive nephron site.

Furosemide activates the renin-angiotensin-aldosterone system (RAAS) and the sympathetic nervous system (SNS). However, postdiuretic Na⁺ retention is not blunted by doses of an ACE inhibitor, which prevents any changes in plasma angiotensin II or aldosterone concentrations,^{248,251,252} or by prazosin, which blocks adrenergic receptors, even when an ACE inhibitor and prazosin are given in combination.²⁴⁹

Micropuncture studies have shown that the blunted natriuretic response to furosemide during repeated administration can be attributed to three factors: (1) reduced NaCl delivery to the site of furosemide action; (2) limited inhibition of NaCl reabsorption by furosemide in the loop of Henle; and (3) enhanced ability of the distal tubule to reabsorb the extra NaCl load delivered during furosemide's upstream action.²⁵³ A recent clinical study has suggested that the third factor is the primary driver for loop diuretic resistance in acute heart failure.²⁵⁴

Rats receiving prolonged infusions of loop diuretics have considerable structural hypertrophy of the DCT, connecting tubule, and intercalated cells of the CD²⁵⁵ that is partially dependent on angiotensin II.²⁵⁶ The DCT and CD have a large increase in mRNA for insulin-like growth factor-binding protein-1²⁵⁷ and increases in both Na⁺-K⁺-ATPase²⁵⁸ and H⁺-ATPase.²⁵⁹ The Na⁺-K⁺-ATPase activity of rat cortical CD segments increases abruptly following an increase in cellular [Na⁺], owing to the mobilization of a latent pool of enzyme.²⁶⁰ There is doubling of NCC expression in the distal tubules of rats adapted to diuretics.²⁶¹ Micropuncture studies of rats adapted to prolonged diuretic infusion have shown enhanced, aldosterone-independent distal Na⁺ and Cl⁻ absorption and K⁺ secretion.²⁶² Therefore, diuretics induce structural and functional adaptations of downstream nephron segments, apparently in response to increased rates of NaCl delivery and, to some extent, to RAAS activation. Nephron adaptation could underlie the inappropriate renal Na⁺ retention that can persist for up to 2 weeks after abrupt cessation of diuretic therapy.²⁶³

Normal subjects fully eliminate a modest NaCl load (100 mmol) over 2 days.²⁶⁴ However, when these subjects are challenged with the same NaCl load delivered after the administration of bumetanide during simultaneous infusion of sufficient fluid, Na⁺, K⁺, and Cl⁻ to prevent any losses, elimination of the load is prevented.²⁶⁴ Thus, diuretics can entrain an ECV-independent NaCl retention; this is apparent when distal delivery is enhanced, as during high NaCl intake. Even a single dose of loop diuretic can cause a Cl⁻ depletion “contraction” alkalosis, which may contribute to diuretic

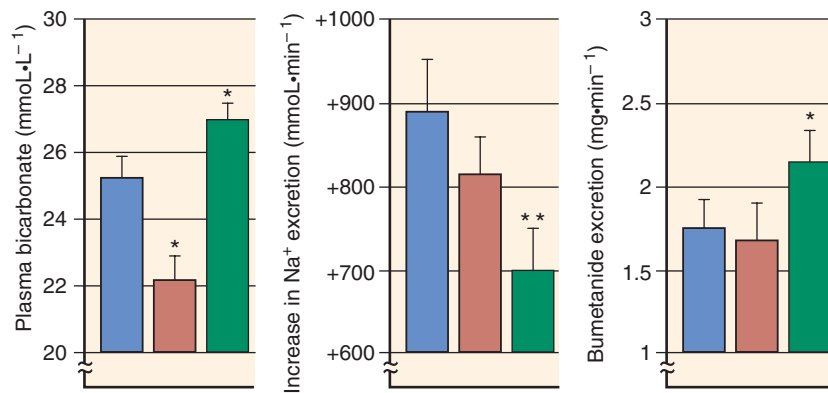


Fig. 50.11 Mean $\pm$ standard error of the mean values for plasma bicarbonate concentration, increase in Na^+ excretion with bumetanide (1 mg intravenously), and rate of bumetanide excretion in normal subjects ($n = 8$) after equilibration to equivalent diets containing 100 mmol/24 hours of NaCl (control, blue bars), NH_4Cl (mild metabolic acidosis, pink bars), or NaHCO_3 (mild metabolic alkalosis, green bars). Compared with control: * $P < .05$; ** $P < .01$. (Modified from Loon NR, Wilcox CS. Mild metabolic alkalosis impairs the natriuretic response to bumetanide in normal human subjects. *Clin Sci [Colch]*. 1998;94:287–292.)

Box 50.1 Strategies to Overcome Diuretic Braking

1. Restrict dietary salt to prevent postdiuretic salt retention.
2. Consider adding another class of diuretic.
3. Consider multiple daily dosing or a diuretic with prolonged action.
4. Do not stop diuretic therapy abruptly.
5. Prevent or reverse diuretic-induced metabolic alkalosis.

tolerance and the braking phenomenon.⁷⁰ In a study of normal subjects in whom mild metabolic alkalosis was produced by the equimolar substitution of NaHCO_3 for NaCl, bumetanide-induced natriuresis was reduced during alkalosis, despite enhanced delivery of bumetanide to the urine (Fig. 50.11).¹²³ This finding implies a profound defect in tubular responsiveness to the diuretic. Several mechanisms may contribute:

1. The Na-K-2Cl cotransporter has affinities for Na^+ , K^+ , and Cl^- of 7.0, 1.3, and 67 mM, respectively. Thus, the $[\text{Cl}^-]$ of tubular fluid may be low enough during Cl^- depletion alkalosis to limit reabsorption by this transporter and thereby limit the responsiveness to loop diuretics.
2. Alkalosis causes glycosylation of the NKCC2, which could alter its transport function.⁶¹
3. The thiazide-sensitive NCC in the rat DCT is increased by 40% during NaHCO_3 administration.²⁶⁵

Findings from these studies have several clinical implications (Box 50.1):

1. Dietary salt intake must be restricted, even in subjects receiving powerful loop diuretics, to obviate postdiuretic salt retention and ensure the development of a negative NaCl balance.
2. During prolonged diuretic administration, subjects may be particularly responsive to another class of diuretic.
3. Diuretic therapy should not be stopped abruptly unless dietary salt intake is curtailed because the adaptive

mechanisms limiting salt excretion persist for days after diuretic use.

4. Selection of a diuretic with a prolonged action, or more frequent administration of the diuretic, will enhance NaCl loss by limiting the time available for postdiuretic salt retention. Indeed, a continuous infusion of a loop diuretic is somewhat more effective than the same dose given as a bolus injection in volunteers²⁶⁶ and in patients with CKD,²⁶⁷ despite a similar delivery of diuretic to the urine. Although a previous study has shown that a continuous infusion is also more effective in cardiac disease,²⁶⁸ a later study showed similar efficacy.²⁶⁹
5. Prevention or reversal of diuretic-induced metabolic alkalosis may enhance diuretic efficacy.

There are similar patterns of furosemide-induced K^+ loss followed by renal K^+ retention²⁷⁰ associated with an increase in the transtubular K^+ gradient.²⁷¹ In contrast, loop diuretics induce ongoing renal K^+ losses during severe salt restriction due to hyperaldosteronism,²⁷⁰ which can be countered by distal, K^+ -sparing diuretics.²⁷¹

HUMORAL AND NEURONAL MODULATORS OF THE RESPONSE TO DIURETICS

RENIN-ANGIOTENSIN-ALDOSTERONE SYSTEM

Diuretic therapy increases plasma renin activity and serum aldosterone concentrations, as described previously. The initial rise in plasma renin activity with loop diuretics is independent of volume depletion or the SNS and is related to the inhibition of NaCl reabsorption at the macula densa.²⁷² Loop diuretics also stimulate renal prostacyclin release, which promotes renin secretion.²⁷³ In longer term use of loop diuretics, renin secretion depends on ECV depletion and the SNS.

Activation of the RAAS in patients treated with diuretics and salt restriction for edema limits the natriuresis.²⁷⁴ In a study of patients with heart failure (HF), ACE inhibition potentiated the diuretic and natriuretic responses to furosemide, despite a drop in blood pressure.²⁷⁵ However, severe volume depletion and azotemia can complicate overzealous

therapy with ACE inhibitors, particularly in patients with HF who are receiving high doses of diuretics or in those with stenosis of both renal arteries or the artery to a single or dominant kidney.²⁷⁶ Thus, the combination of diuretics and ACE inhibitors can be highly effective but requires careful surveillance.

During stimulation of the RAAS by severe dietary salt restriction, further diuretic-induced increases in serum aldosterone concentration promote renal K^+ losses.²⁷⁷ ACE inhibitors counter diuretic-induced increases in serum aldosterone concentration and blunt diuretic-induced hypokalemia.²⁷⁰

EICOSANOIDS

PGE₂ acting on EP₄ inhibits NaCl reabsorption via the NKCC2²⁷⁸ and inhibits free water and Na^+ reabsorption in the CDs via changes in cAMP (Fig. 50.12).²⁷⁹ Loop diuretics, thiazides, triamterene, and spironolactone increase prostaglandins substantially.²⁸⁰ Inhibition of PG synthesis by NSAIDs can diminish the natriuresis and diuresis induced by furosemide,²⁸¹ hydrochlorothiazide,²⁸² spironolactone,²⁸⁰ or triamterene (see Fig. 50.12).²⁸³ Microperfusion of the loop segment with PGE₂¹²⁹ restores the response to furosemide in indomethacin-treated rats. Indomethacin also blunts furosemide-induced renal²⁸⁴ and capacitance vessel vasodilation²⁸⁵ and stimulation of renin.²⁸⁶ The blunting of furosemide-induced natriuresis by NSAIDs is potentiated by salt depletion²⁸⁷ and is prominent in edematous patients.²⁸¹ The NSAIDs ibuprofen, naproxen, and sulindac similarly blunt furosemide-induced natriuresis. A COX-2 inhibitor blocks furosemide-induced renin secretion but not natriuresis,

although other studies have shown an effect of COX-2 inhibition on Na^+ reabsorption.²⁸⁸ It is COX-1 that facilitates natriuresis in the distal nephron.²⁸⁹ Salt-sensitive hypertensive subjects have a blunted natriuretic response to furosemide that may be related to a paradoxical reduction in the renal excretion of 20-HETE. Thus, COX-1 products mediate a part of furosemide-induced natriuresis, whereas COX-2 products mediate renin secretion. 20-HETE may be an important positive modulator of salt excretion.

Loop diuretics also increase the excretion of the thromboxane A₂ (TXA₂) metabolite TxB₂. Inhibition of TXA₂ synthesis or receptors in the rat increases furosemide diuresis²⁹⁰ and diminishes the renal vasodilation.²⁹¹ Thus, TxA₂ may antagonize the actions of loop diuretics.

ARGININE VASOPRESSIN

AVP levels increase after the administration of furosemide.²⁹² This may be a response to a reduced blood volume. Plasma AVP levels are increased in many edematous states, such as HF and liver cirrhosis,²⁹³ especially in those who demonstrate hyponatremia during thiazide treatment.²⁹⁴ AVP stimulates K^+ secretion in the rat distal tubule.²⁹⁵ Diuretic-induced AVP release contributes to hypokalemia because the kaliuretic response to furosemide is reduced by 40% in subjects whose AVP release is suppressed by a water load.²⁷⁷ In addition, despite its classic role as a water balance hormone, AVP has been shown to increase the activity of both the NCC and ENaC, possibly aggravating sodium retention in edematous states.^{296,297}

CATECHOLAMINES AND SYMPATHETIC NERVOUS SYSTEM

The first dose of furosemide raises the heart rate and plasma catecholamine concentrations.^{123,249} Blockade of α_2 -adrenergic receptors with prazosin does not modify the ensuing renal salt retention²⁴⁹ but blockade of these receptors blunts the renin release.²⁷² Short-term, furosemide-induced ECV depletion in the conscious rat activates sympathetic nerve activity that stabilizes the blood pressure.²⁹⁸

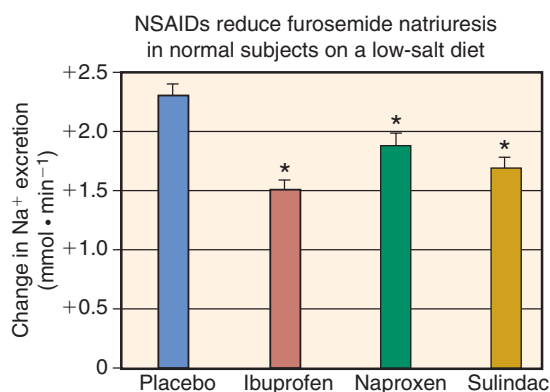
ATRIAL NATRIURETIC PEPTIDE

Diuretics are often used to treat patients who have an expanded blood volume and elevated levels of ANP. Administration of furosemide to dogs with heart failure reduces ANP levels.²⁹⁹ Infusion of ANP in this dog model promotes furosemide-induced natriuresis and blunts activation of the RAAS and the fall in the GFR. Thus, a drop in ANP levels contributes to postdiuretic renal NaCl retention.²⁹⁹

DIURETIC RESISTANCE

Diuretic resistance implies an inadequate clearance of edema, despite a full dose of diuretic.³⁰⁰ The principal causes are summarized in Box 50.2. The first step in addressing inadequate diuretic response is to select the appropriate target response (e.g., a specific body weight) and to ensure that the edema is due to inappropriate renal NaCl and fluid retention, rather than to lymphatic or venous obstruction or redistribution (Fig. 50.13). Diuretics do not prevent edema caused by dihydropyridine calcium channel blockers.³⁰¹

The next step is to exclude poor adherence, severe blood volume depletion, and concurrent NSAID use. Thereafter,



Mean $\pm$ SEM (n=11) studied in crossover design after equilibration to 10 mmol 24h⁻¹ Na^+ diet

Response to 40 mg of furosemide IV during:

■ Placebo
■ Ibuprofen 600 mg per 6 hr for 3 doses
■ Naproxen 375 mg per 12 hr for 2 doses
■ Sulindac 200 mg per 12 hr for 2 doses

Compared to placebo: *, $P < 0.05$ vs placebo

Fig. 50.12 Mean $\pm$ standard error of the mean (SEM) values for change in Na^+ excretion for 11 normal subjects given 40 mg of furosemide intravenously after placebo or after each of three nonsteroidal antiinflammatory drugs: ibuprofen, 600 mg/6 hr for three doses; naproxen, 375 mg/12 hr for two doses; or sulindac, 200 mg/12 hr for two doses. NSAIDs, Nonsteroidal antiinflammatory drugs. (Modified from Brater DC, Anderson S, Baird B, et al. Effects of ibuprofen, naproxen, and sulindac on prostaglandins in men. *Kidney Int.* 1985;27:66–73.)

Box 50.2 Common Causes of Diuretic Resistance

Incorrect diagnosis (e.g., venous or lymphatic edema)
Inappropriate NaCl or fluid intake
Inadequate drug reaching tubule lumen in active form because of:

- Noncompliance
- Dose inadequate or too infrequent
- Poor absorption (e.g., due to uncompensated HF)
- Decreased renal blood flow (e.g., due to HF or cirrhosis of liver, or in older patient)
- Decreased functional renal mass (e.g., due to AKI or CKD, or in older patient)
- Proteinuria (e.g., due to nephrotic syndrome)

Inadequate renal response because of:

- Low glomerular filtration rate (e.g., due to AKI, CKD)
- Decreased effective extracellular fluid volume (e.g., due to edematous conditions)
- Activation of renin angiotensin aldosterone system (e.g., due to edematous conditions)
- Nephron adaptation (e.g., due to prolonged diuretic therapy)
- Nonsteroidal antiinflammatory drugs (e.g., indomethacin, aspirin)

AKI, Acute renal failure; HF, heart failure; CKD, chronic kidney disease.

dietary NaCl intake should be quantitated. In the steady state, this can be assessed from measurements of 24-hour Na^+ excretion. For patients with mild edema or hypertension, a daily Na^+ intake of 100 to 120 mmol may be sufficient. For patients with diuretic resistance, the help of a dietitian is usually necessary to reduce daily Na^+ intake to 80 to 100 mmol. In patients who are not in a steady state (e.g., patients with worsening heart failure), urine sodium levels will be low, and this observation may be used as an indication to intensify diuretic therapy.³⁰² The diuretic dose must be above the natriuretic threshold (the steep part of the dose-response curve in Fig. 50.8). Outpatients should be able to detect an increase in urinary volume within 4 hours of an administered dose; urine volume or body weight can be measured directly in patients who are hospitalized. If a diuresis does not occur, the next step is to double the dose until an effective dose or the maximum safe dose is reached.

The next step is to give two daily doses of the diuretic. Furosemide and bumetanide act for only 3 to 6 hours. Two daily half-doses, by interrupting postdiuretic salt retention, produce a greater response than the same total dose given once daily, as long as both are above the diuretic threshold. Concurrent disease may impair absorption of the diuretic. Thus, a more bioavailable diuretic, such as torsemide, may be preferable to furosemide.⁹⁸ Diuretic resistance is often accompanied by a pronounced metabolic alkalosis,¹²³ which may be reversed by the administration of KCl or addition of a distal K^+ -sparing diuretic.

A progressive increase in diuretic dosage may produce an inadequate reduction in body fluids because of activation of NaCl-retaining mechanisms. ACE inhibitors can sometimes restore a diuresis in resistant patients with HF,²⁷⁴ but a fall in blood pressure often limits the response. Adaptive changes

in downstream nephron segments during prolonged diuretic therapy^{83,255} provide a rational basis for combining diuretics (see the following section). Highly resistant patients can be admitted for a trial of intravenous infusion of a loop diuretic or ultrafiltration.³⁰³

DIURETIC COMBINATIONS

Full doses of more than one diuretic acting on the same transport mechanism are less than additive, whereas use of several diuretics acting on a separate mechanism may be synergistic.^{5,304}

LOOP DIURETICS AND THIAZIDES

A loop diuretic and a thiazide or thiazide-like drug (e.g., hydrochlorothiazide, metolazone) are synergistic in normal subjects and in subjects with edema or renal insufficiency.^{304–308} Metolazone is equivalent to bendroflumazide in enhancing NaCl and fluid losses in furosemide-resistant subjects with HF or the nephrotic syndrome.³⁰⁹ During prolonged furosemide therapy, the responsiveness to a thiazide is augmented.⁸³ Patients with advanced CKD ($\text{eGFR} < 30 \text{ mL/min/1.73 m}^2$) that is unresponsive to a thiazide alone show a marked natriuresis when a thiazide is added to loop diuretic therapy,³⁰⁷ probably by blockade of enhanced distal tubular Na^+ reabsorption.³¹⁰ However, such combination therapy should be initiated under close surveillance because of a high associated incidence of hypokalemia, excessive ECV depletion, and azotemia.³¹¹

LOOP DIURETICS OR THIAZIDES AND DISTAL POTASSIUM-SPARING DIURETICS

Amiloride or triamterene increases furosemide natriuresis only modestly but curtails the excretion of K^+ and net acid⁶⁸ and preserves total body K^+ .³¹² Distal K^+ -sparing agents are generally contraindicated in renal failure because they may cause severe hyperkalemia and acidosis.

CLINICAL USES OF DIURETICS

A general algorithm for diuretic therapy in the treatment of CKD, nephrotic syndrome, liver cirrhosis, and HF is shown in Fig. 50.14.³¹³

EDEMATOUS CONDITIONS

The first aim in the treatment of edema is to reverse the primary cause by restoring hemodynamics and cardiac output in patients with heart failure (e.g., use of vasodilators or elimination of cardiac depressant drugs), by improving hepatic function in patients with cirrhosis and ascites (e.g., stopping alcohol intake), or by diminishing proteinuria in patients with the nephrotic syndrome (e.g., administration of ACE inhibitors or ARBs). Although the GFR is not reduced by low-dose diuretic therapy in normal subjects, it can be reduced in those with CKD or if there is an abrupt fall in blood pressure (BP), especially if this is complicated by orthostatic hypotension. Moreover, overzealous diuresis decreases the cardiac output, BP, and renal function, and stimulates the RAAS, SNS, prostaglandins, and AVP, all of which may compromise the desired hemodynamic and renal responses.³¹⁴ Therefore, diuretic therapy for edema should be initiated

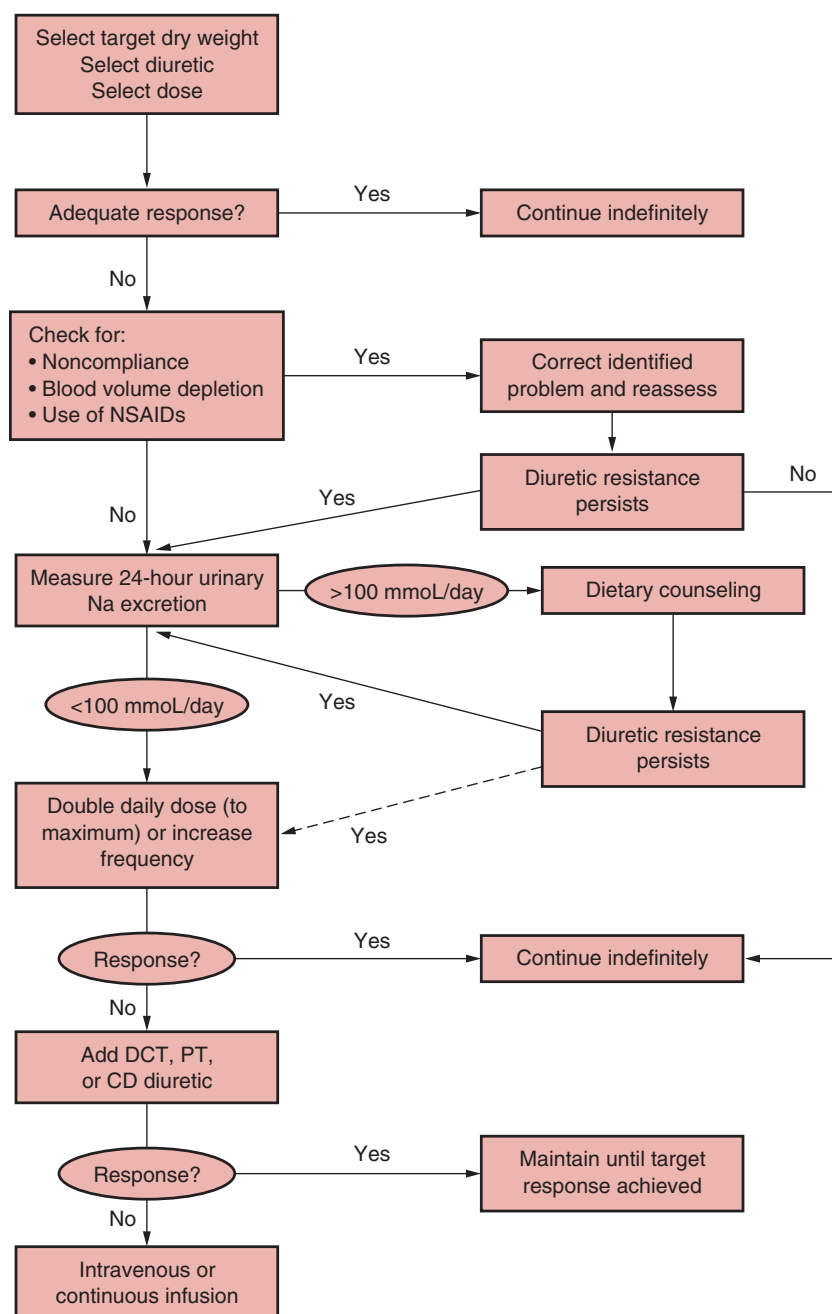


Fig. 50.13 Diagrammatic representation of an approach to the management of a patient with resistance to a loop diuretic. *CD*, Collecting duct diuretic (e.g., amiloride, triamterene, or spironolactone); *DCT*, distal convoluted tubule diuretic (e.g., thiazide); *NSAIDs*, nonsteroidal antiinflammatory drugs; *PT*, proximal tubule diuretic (e.g., acetazolamide). (From Ellison DH, Wilcox CS. Diuretics: use in edema and the problem of resistance. In Brady HR, Wilcox CS, eds. *Therapy in Nephrology and Hypertension*. ed 2. London: Elsevier Science; 2003.)

with the lowest effective dose.³¹⁵ Additional drugs can be used to counteract unwanted actions. For example, ACE inhibitors, ARBs, or MRAs can prevent RAAS activation and enhance fluid losses, yet diminish K^+ depletion (see Fig. 50.20). The use of a second diuretic can have a synergistic action, whereas the use of a distal K^+ -sparing agent may counteract unwanted hypokalemia, alkalosis, or Mg^{2+} depletion (see “Adaptation to Diuretic Therapy”).

Dietary Na^+ intake should be restricted to 2.5 to 3 g daily (corresponding to 107–129 mmol/24 hours) in patients with

mild edema. Increasingly severe Na^+ restrictions to 2 g daily (86 mmol/24 hours) are required for patients with refractory edema.

Some resistance to diuretic therapy should be anticipated in all patients with CKD and those with more than mild edema (Fig. 50.15).

HEART FAILURE

HF is classified as occurring with reduced or preserved ejection fraction, yet both forms of HF require diuretics as cornerstones

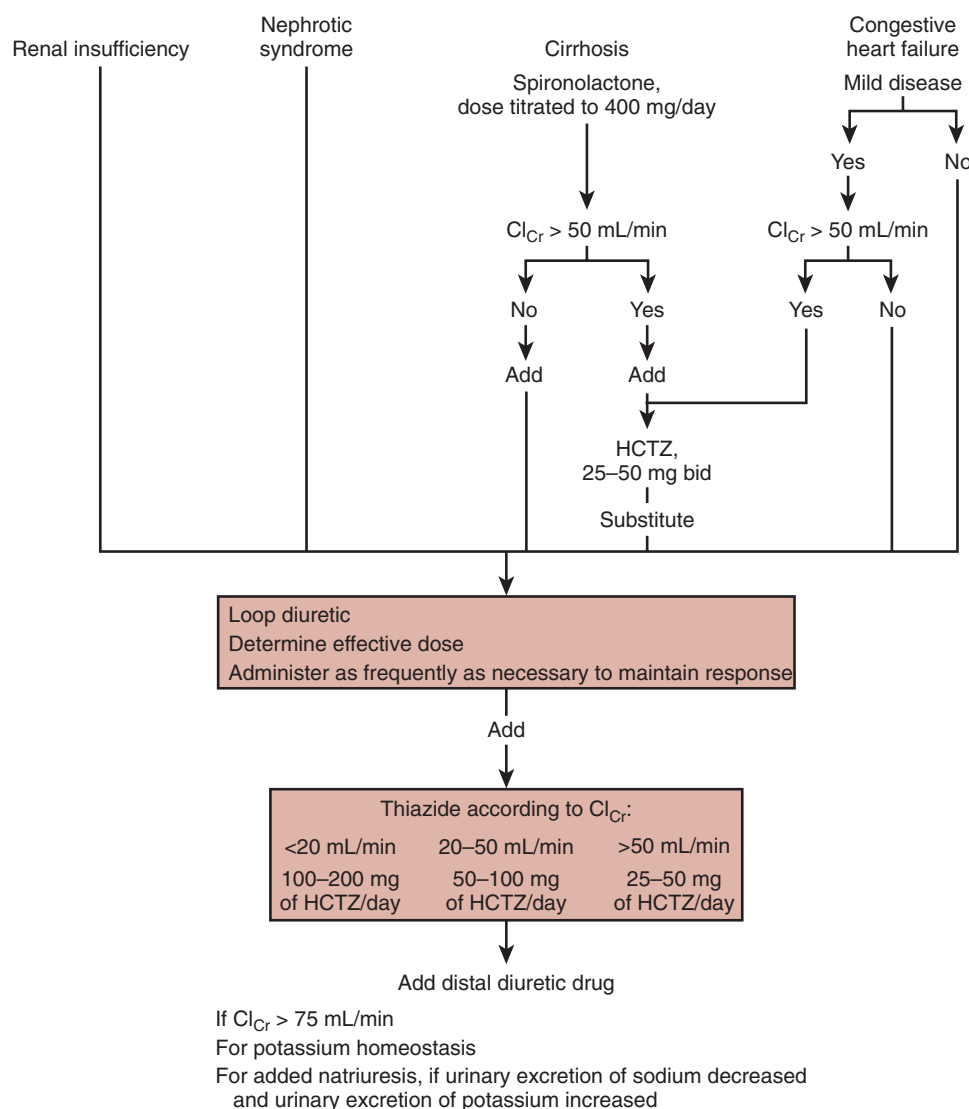


Fig. 50.14 Algorithm for diuretic therapy in patients with edema due to renal, hepatic, or cardiac disease. *bid*, Twice a day; Cl_{Cr} , creatinine clearance; *HCTZ*, hydrochlorothiazide. (From Brater DC. Diuretic therapy. *N Engl J Med*. 1989;339:387–395.)

of therapy. The approach to cardiac failure depends on the cause and whether there is acute decompensation or a compensated chronic state.^{316,317} This section first reviews the role of diuretics in acute decompensated HF in general and as secondary to acute coronary syndrome and then discusses the role of diuretics as maintenance therapy in systolic and diastolic HF. The reader is also referred to a recent review³¹⁸ and to the joint guidelines published by the American College of Cardiology and American Heart Association for more detailed recommendations and an overview of the level of evidence for each treatment.³¹⁹

Acute Decompensated Heart Failure

In the absence of obvious causes such as acute coronary syndrome and valve abnormalities, acute decompensated HF (ADHF) often results from an imbalance in the neurohumoral systems that regulate cardiac and renal function.³²⁰ Therefore, it is rational to target these mechanisms with selective therapy. After initial stabilization, the mainstay of treatment is vasodila-

tor and diuretic therapy. Intravenous vasodilators such as nitroglycerine, nitroprusside, and nesiritide counteract the effects of baroreceptor-dependent increases in sympathetic tone, angiotensin II and aldosterone, endothelin, and AVP. Vasodilators are usually combined with intravenous loop diuretics (e.g., furosemide, 40 mg; bumetanide, 1 mg; torsemide, 10–20 mg). For example, a study in patients with severe HF has shown that therapy with vasodilators and diuretics aimed at improving overall hemodynamic status leads to rapid neurohumoral improvement when central filling pressure declined.³²¹ Another study has found that although “aggressive decongestion” during ADHF is associated with worsening renal function, survival is actually improved.³²²

In a carefully performed randomized trial, no differences between continuous dosing and bolus dosing of furosemide were observed; furthermore, no differences in global symptom improvement were observed between a low dose (equivalent of an oral dose) and a high dose (2.5 times the oral dose).²⁶⁹ It should be noted, however, that this trial did not study

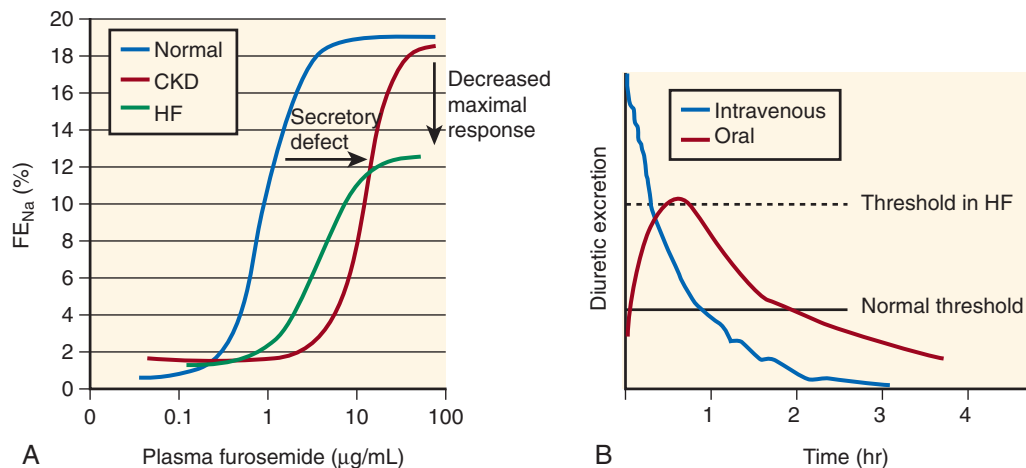


Fig. 50.15 Dose-response curves for loop diuretics. (A) The fractional Na^+ excretion (FE_{Na}) as a function of plasma loop diuretic concentration. Compared with normal subjects, patients with chronic kidney disease (CKD) show a rightward shift in the curve owing to impaired diuretic secretion. The maximal response is preserved when expressed as FE_{Na} , but not when expressed as absolute Na^+ excretion. Patients with congestive heart failure (HF) demonstrate a rightward and downward shift, even when expressed as FE_{Na} , and thus are relatively diuretic-resistant. (B) Comparison of the response to intravenous and oral doses of loop diuretics. In a normal individual, an oral dose may be as effective as an intravenous dose because the time during which this individual is above the natriuretic threshold (indicated by the *Normal threshold* line) are approximately equal. If the natriuretic threshold increases (as indicated by the *Threshold in HF* line), then the oral dose may not provide a high enough plasma level to elicit natriuresis.

patients shown to be diuretic-resistant. Longer acting loop diuretics, such as torsemide³²³ and azosemide,³²⁴ produce less neurohumoral activation and may be preferable.

Diuretic kinetics are impaired in decompensated HF.^{325,326} The bioavailability of furosemide, unlike that of bumetanide or torsemide, is erratic in HF.³²⁷ This feature, and a longer duration of action, may account for the finding of a 50% reduction in the requirement for readmission to hospital in patients with HF randomly assigned to receive torsemide rather than furosemide.⁹⁸ There is decreased plasma clearance in decompensated HF because of a decreased RBF.³²⁵ Together, these effects can limit the peak diuretic concentration in the tubular fluid to the foot of the dose-response curve and thereby diminish the response (see Fig. 50.15).

Nesiritide is a recombinant human B-type natriuretic peptide (BNP) that was approved by the US Food and Drug Administration (FDA) for the treatment of ADHF.³²⁸ Studies of mice deficient in natriuretic peptide receptor A have implicated this system in the natriuretic response to blood volume expansion³²⁹ and provide a rationale for BNP in the treatment of ADHF. Nesiritide given short term to patients with decompensated HF can reduce pulmonary capillary wedge pressure.³³⁰ However, some subsequent studies have shown that nesiritide is not a diuretic in patients with HF and, in comparison with placebo, may be associated with higher risks of death and worsening renal function.^{331–333} Conversely, other studies have indicated a potential role for nesiritide in the perioperative treatment of patients with left ventricular dysfunction who are undergoing cardiac surgery. In this context, these agents may improve renal function and possibly even survival.^{334,335} Current guidelines recommend nesiritide for alternative vasodilator therapy in patients with ADHF without hypotension and with volume overload who remain dyspneic, despite receiving intravenous loop diuretics.^{319,336}

In the past few years, there has been substantial interest in venovenous ultrafiltration as treatment for ADHF.³³⁷ Nevertheless, a randomized trial conducted by the Heart Failure Clinical Research Network, comparing diuretic therapy with ultrafiltration, found that creatinine levels rose more in the ultrafiltration group than in the diuretic group, despite similar volume losses.³³⁸ It should be emphasized, however, that this trial specifically excluded individuals in whom dialysis, in addition to volume removal, was indicated. The supplementary material in this publication³³⁸ also provides protocols that have been widely promulgated by the Heart Failure Clinical Research Network for volume reduction in decompensated CHF. A recent meta-analysis has shown that ultrafiltration results in better clinical decongestion than intravenous loop diuretics, but does not improve rehospitalization or mortality rates.³³⁹ Furthermore, ultrafiltration may cause serious product-related adverse events.³⁴⁰

Finally, the intravenous inotropes dobutamine and milrinone are reserved for situations in which ADHF is complicated by unresponsiveness to standard therapies, diminished peripheral perfusion, end-organ dysfunction, and/or hypotension (low-output syndrome). One trial has shown no added benefit of low-dose dopamine or nesiritide in decongestion or kidney function.³⁴¹

ACUTE DECOMPENSATED HEART FAILURE IN ACUTE CORONARY SYNDROME

Patients with acute myocardial infarction (AMI) require the rapid establishment of coronary reperfusion (e.g., by thrombolysis and percutaneous coronary intervention) and treatment of arrhythmias. The aim of concomitant treatment is to counter the increase in left ventricular end-diastolic pressure, which enhances wall tension and O_2 usage, and the accumulation of pulmonary edema without further curtailing the cardiac output. Judicious use of diuretics may

meet these requirements. In one study, intravenous furosemide for left ventricular failure (LVF) complicating acute AMI reduced the left ventricular filling pressure from 20 to 15 mm Hg within 5 to 15 minutes and increased the venous capacitance by 50%.³⁴² This rapid venodilation is blocked by NSAIDs²⁸⁵ and ACE inhibitors.³⁴³ The ensuing diuresis reduces left ventricular end-diastolic pressure further.

A study of first-line therapy for 48 patients with acute LVF following AMI compared the responses to intravenous furosemide, a venodilator (isosorbide dinitrate), an arteriolar dilator (hydralazine), and a positive inotrope (prenalterol).³⁴⁴ The venodilator and furosemide both reduced left ventricular filling pressure while maintaining the cardiac index and heart rate. The investigators concluded that these were the best first-line agents but that they should be combined with an arteriolar vasodilator. In contrast, a study randomly assigned 110 patients with acute LVF to receive either high-dose isosorbide dinitrate (3 mg intravenously every 5 minutes) or high-dose furosemide (80 mg intravenously every 15 minutes).³⁴⁵ An adverse end point occurred more frequently in those receiving furosemide (46%) than in those receiving nitrate (25%). The investigators cautioned against the use of high-dose furosemide in acute LVF. Although intravenous furosemide decreases left ventricular filling pressure in patients with LVF, the shape of the Frank-Starling ventricular function curve predicts little change in cardiac output at elevated filling pressures. Nevertheless, most investigators recommend a trial of loop diuretics after ECV depletion and preload-dependent right HF have been ruled out by targeted volume boluses.³⁴⁶

Furosemide can be given as an IV bolus of up to 100 mg or as a short-term infusion to limit the risk of ototoxicity. Although controversial, an intermittent infusion produces a slightly greater natriuresis in most comparative studies.³⁴⁷ Ideally, lower doses are used initially and titrated up to a good effect—for example, a pulmonary capillary wedge pressure of 16 mm Hg.³⁴⁷ Hemodynamic and biochemical parameters should be monitored frequently during loop diuretic therapy, especially because patients with AMI are usually also treated with beta-blockers, ACE inhibitors, or vasodilators.

Chronic Heart Failure

Diuretics are extremely useful in the long-term management of chronic HF with reduced ejection fraction. Avid renal NaCl and fluid retention leads to pulmonary edema that limits ventilation. Cardiac dilation limits cardiac function and increases wall tension and O₂ usage. This combination can create a spiral of decreasing oxygenation and cardiac output. In a study of 13 patients with severe edema due to HF, furosemide therapy increased stroke volume by 15% and decreased peripheral vascular resistance, despite reducing body weight by an average of 10 kg.³⁴⁸ In another study, combined therapy using a diuretic and vasodilator reduced left and right atrial volumes, corrected atrioventricular valvular regurgitation, and improved stroke volume by 64%.³⁴⁴ A meta-analysis of trials for HF has concluded that odds ratios were reduced to 0.25 for mortality and 0.31 for hospitalization for subjects randomly selected to receive diuretics.³⁴⁹ These remarkable data strongly support the use of diuretics in HF.

On the other hand, the failing heart has a decreased capacity to regulate its contractility in response to changes

in venous return, so if diuretic therapy is too abrupt or severe, the patient suffers from a decreased effective blood volume—orthostatic hypotension, weakness, fatigue, decreased exercise ability, and prerenal azotemia. This is especially true for patients with diastolic dysfunction, an increasingly recognized form of symptomatic HF with preserved ejection fraction.³⁵⁰ Therefore, salt-depleting therapy requires continual reassessment and judicious use of other measures (e.g., vasodilators, ACE inhibitors, ARBs, or MRAs). Mild HF often responds to dietary Na⁺ restriction (100–120 mmol/day) and low doses of a thiazide diuretic. As cardiac failure progresses, larger, more frequent doses of loop diuretics and tighter control of dietary salt (80–100 mmol/day) are required. It is important to emphasize that diuretic responsiveness is impaired in patients with advanced HF, as shown by a shift to the right in the natriuresis-excretion relationship of diuretics (see Fig. 50.15).³²⁶ For the refractory patient, the addition of a second diuretic acting at the proximal tubule (e.g., acetazolamide) or a downstream site (e.g., a thiazide) can produce a dramatic diuresis, even in individuals with impaired renal function.^{351,352} For example, additional therapy with metolazone³⁵³ or hydrochlorothiazide³⁵⁴ can increase fluid losses by an average of 7 to 8 kg. By comparing the fractional lithium and sodium excretion after intravenous loop diuretics, a recent study has confirmed that distal tubular compensatory sodium reabsorption is the primary driver of diuretic resistance in heart failure.³⁵⁵

Although drug therapy for chronic HF should be individualized, a suggested step-up algorithm consists of diuretics and ACE inhibitors or ARBs as the first agents, beta-blockers second, and MRAs or hydralazine-isosorbide dinitrate (in blacks) third, with digoxin, cardioverter-defibrillator, and resynchronization devices being saved as a final resort.³⁵⁶ The lesson from the excess morbidity from hyperkalemia observed during trials of spironolactone therapy is to use MRAs within guidelines.²¹¹ The guidelines exclude men with serum creatinine concentrations greater than 2.5 mg/dL (221 μmol/L) and women with serum creatinine concentrations greater than 2.0 mg/dL (177 μmol/L).³⁵⁷ However, within guidelines, both MRAs and loop diuretics are important because they can improve ventricular remodeling.^{358,359} In patients with preserved ejection fraction, spironolactone may improve left ventricular diastolic dysfunction, but it has been found not to improve symptoms and outcome.^{360,361}

Despite these options, HF often progresses and introduces a vicious cycle. Decompensated HF stimulates the RAAS and AVP,³⁶² predisposing to hypokalemia, hypomagnesemia, hyponatremia, and arrhythmias. Hypokalemia potentiates the binding of digitalis to cardiac myocytes,³⁶³ decreases its renal elimination,³⁶⁴ and enhances its cardiac toxicity.³⁶⁵ Although this cycle can be curtailed by higher doses of diuretics or continuous intravenous infusion of a loop diuretic,³⁶⁶ the risks of volume depletion, azotemia, and electrolyte abnormalities increase sharply.³⁶⁷ Continuous infusion should be tried in patients known to show a response to maximum bolus doses of diuretics because bolus therapy results in higher initial serum concentrations and therefore higher initial rates of urinary diuretic excretion than continuous infusion. If the patient has received one or more intravenous boluses within the previous few hours, then an infusion can be started without a loading dose.

Furosemide infusion rates of up to 240 mg/hour have been reported in the literature. The risk of ototoxicity and other side effects associated with these infusion rates must be weighed against alternative strategies, such as the addition of a thiazide-type diuretic or fluid removal via ultrafiltration.

Therefore, new therapies are required because a decrement in renal function predicts a bad outcome in patients treated for HF.³⁶⁸ Renal dysfunction can be ameliorated by an ARB, provided that BP is maintained. The combination of an ARB with a neprilysin inhibitor, which prevents the breakdown of natriuretic peptides, may be especially advantageous in terms of cardiovascular and renal outcomes.^{233,369} In the scenario in which deteriorating HF worsens kidney function, often referred to as the *cardiorenal syndrome*, therapy must be aimed at improving cardiac function (if possible) and treating congestion with diuretics, beta-blockers, ACE inhibitors, or ARBs and MRAs.³⁷⁰

Right Ventricular Failure

The requirement for diuretic therapy in patients with pure right HF or cor pulmonale is not compelling. A decrease in venous return induced by vigorous diuresis may worsen right heart function. Furosemide administration increases angiotensin II-induced hypoxic pulmonary vascular resistance.³⁶⁵ Therefore, the emphasis should be on the reversal of chronic hypoxemia.

CIRRHOSIS OF THE LIVER

Most patients with cirrhotic ascites and peripheral edema have expansion of the ECV owing to arteriolar underfilling, which is caused by peripheral vasodilation and impaired cardiac function.^{371–373} Studies in patients with cirrhosis have also demonstrated increases in proximal reabsorption in response to a diminished effective arterial blood volume.³⁷⁴ Finally, patients with liver cirrhosis have increases in both the natriuretic response to a thiazide and serum aldosterone concentrations.³⁷⁵ Thus the use of diuretics acting on the distal nephron and MRAs are rational for cirrhosis and are usually well tolerated.¹⁷⁹ The American Association for the Study of Liver Disease practice guidelines have suggested that first-line treatment of patients with cirrhosis and ascites should consist of sodium restriction (2000 mg, or 88 mmol daily) and diuretics (oral spironolactone, with or without oral furosemide).³⁷⁶ The guideline suggests an initial regimen of 40 mg furosemide and 100 mg spironolactone, with titration upward maintaining the same diuretic ratio.³⁷⁶ Maximally recommended doses are 400 mg of spironolactone and 160 mg of furosemide.

Patients with cirrhosis and ascites cannot usually tolerate ACE inhibitors or ARBs because of a fall in BP.³⁷⁷ Mild edema without ascites can be treated with dietary restriction of Na⁺ (100 mmol/day). Dietary fluid restriction is not necessary in most patients with cirrhotic ascites. The guideline does not recommend fluid restriction in patients with cirrhosis and ascites, unless the serum sodium level falls below 125 mmol/L.³⁷⁶ Although it has never been shown that treating hyponatremia in liver disease improves prognosis, hyponatremia by itself has been a consistent predictor of poor prognosis.³⁷⁸ Ascitic fluid is largely cleared by the lymphatics. Diuretics increase thoracic duct lymph flow.³⁷⁹ Thus, diuretics decrease ascites formation by decreasing

venous and portal hydraulic pressures, concentrating the plasma proteins,³⁸⁰ and increasing ascites absorption.^{379,381}

The maximal daily ascites drainage into the systemic circulation is limited to 300 to 900 mL.³⁸² Therefore, the maximum daily weight loss in nonedematous patients should not exceed 0.3 to 0.5 kg. In patients with ascites and edema, daily diuretic-induced weight losses of 1 to 3 kg do not perturb the plasma volume or renal function.³⁸³ The same diuretic regimen maintained after the peripheral edema has cleared, however, or given to nonedematous patients, reduces plasma volume by as much as 24% and raises the risks of hyponatremia, alkalosis, and azotemia. Furthermore, the reduced serum albumin level and an increased portal venous pressure, coupled with preexisting diuretic use, can lead to true “underfill edema.” Diuretic therapy for patients with these findings is complicated by hypotension, azotemia, and electrolyte dysfunction. Thus, a diuretic prescription that is initially safe must be reviewed continuously. In addition, patients with ascites but without peripheral edema seem more prone to the development of the side effects of diuretics.³⁸³

The most common problems with furosemide in cirrhosis are electrolyte disturbances and volume depletion. Hypokalemia, which is related to preexisting K⁺ depletion and hyperaldosteronism, can be countered with the use of spironolactone, eplerenone, or a distal K⁺-sparing agent, as noted previously. However, hyperkalemic metabolic acidosis can develop in patients with cirrhosis who are given spironolactone.³⁸⁴ Most severe diuretic resistance requires paracentesis. It is important, however, to differentiate diuretic resistance from poor adherence to NaCl restriction. This can be done by determining 24-hour NaCl excretion or by using a spot urine Na/K ratio; higher ratios suggest poor adherence and lower ratios diuretic resistance; the optimal cutoff varies among studies, between 1 and 2.5.³⁸⁵ Controlled trials in patients with refractory ascites have shown that large-volume paracentesis is more effective than diuretic therapy in reducing hospital stay and electrolyte complications but does not influence mortality.³⁸⁶ Even repeated, large-volume paracenteses (4–6 L/day) are safe if intravenous albumin (40 g with each procedure) is administered.³⁸⁶ Most investigators, however, recommend paracentesis only for cases that are relatively resistant to diuretics and dietary Na⁺ restriction (Fig. 50.16).³⁸⁷

Patients with mild cirrhosis of the liver have a normal or reduced natriuretic response to furosemide, with little change in diuretic kinetics.² However, in those with advanced disease, furosemide absorption is slowed,³⁸⁸ its volume of distribution is increased because of hypoalbuminemia and an expanded ECV, and its elimination is delayed because of hypoalbuminemia, which limits proximal tubule diuretic secretion, and a low RBF, which limits renal clearance.² Resistance to loop diuretics in early cirrhosis is largely due to decreased responsiveness to the drug, which correlates with elevated serum aldosterone levels.¹⁷⁹ With the development of ascites, a further decrease in natriuretic response correlates with decreased delivery of furosemide to the urine³⁸⁹ and with further stimulation of the RAAS.¹⁷⁹

Diuretic resistance is common in advanced cirrhosis. In addition to the usual causes (see Box 50.2), it may herald the development of infection, bleeding, or a critical drop in cardiac output. Patients whose disease is refractory and who

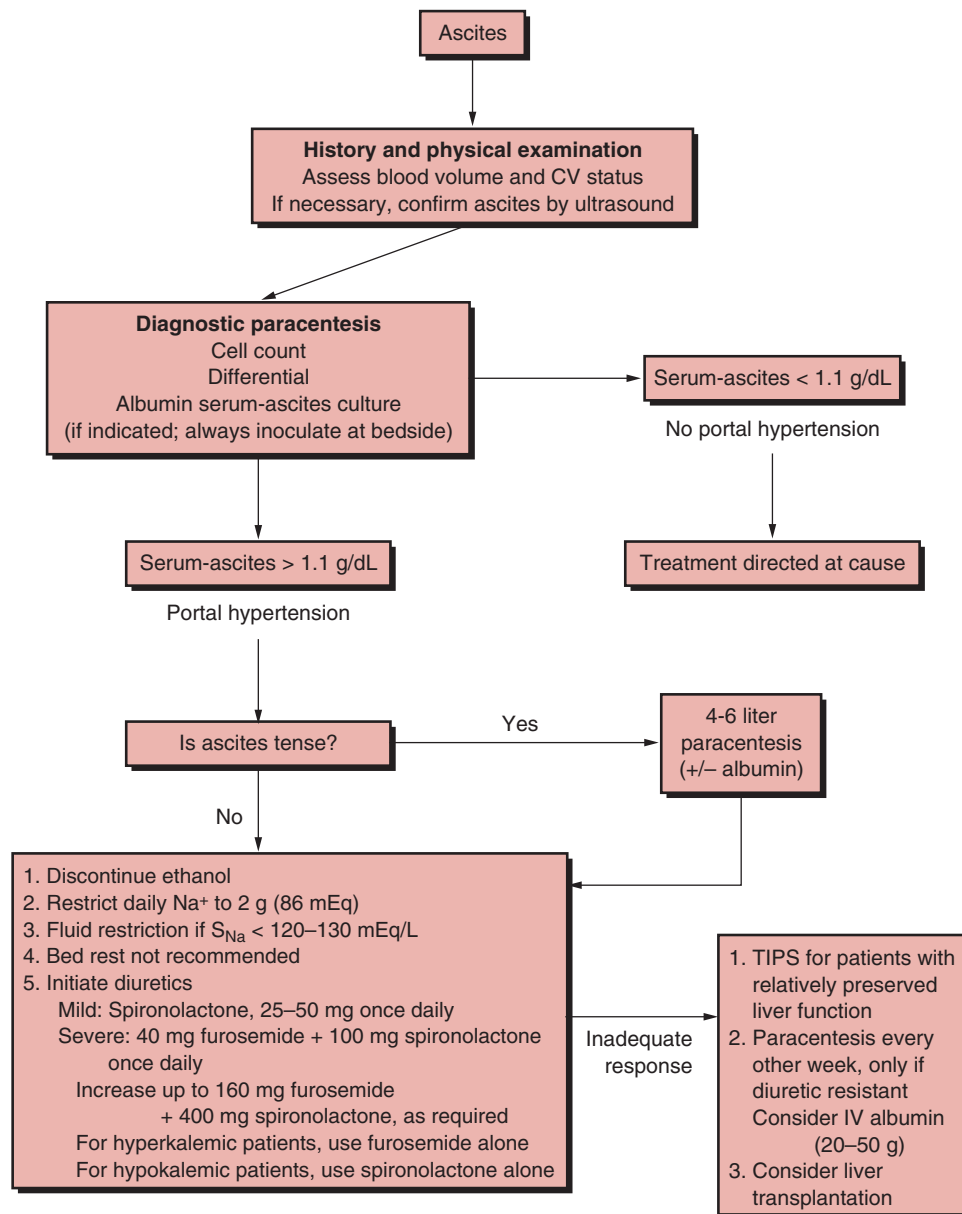


Fig. 50.16 Treatment algorithm for the management of fluid retention in patients with hepatic cirrhosis and ascites. CV, Cardiovascular; IV, intravenous; S_{Na} , serum Na^+ concentration; TIPS, transjugular intrahepatic portosystemic shunt. (From Ellison DH, Wilcox CS. Diuretics: use in edema and the problem of resistance. In Brady HR, Wilcox CS, eds. *Therapy in Nephrology and Hypertension*. ed 2. London: Elsevier Science; 2003.)

are disabled by recurrent paracentesis may show response to body compression³⁹⁰ or a transjugular intrahepatic portosystemic shunt.³⁹¹ Intravenous loop diuretics are generally discouraged because they may precipitate the hepatorenal syndrome.

NEPHROTIC SYNDROME

Urinary albumin losses and reduced hepatic synthesis in the nephrotic syndrome eventually lead to hypoalbuminemia. The ensuing fall in plasma oncotic pressure increases the flux of fluid into the interstitial spaces, leading to underfill edema.^{392,393} Additionally, a primary renal salt retention can lead to overfill edema (Fig. 50.17). Patients with minimal change disease often have contracted plasma volume and stimulated RAAS, whereas those with other causes

of nephrotic syndrome usually have an expanded plasma volume and suppressed RAAS.³⁹⁴ Micropuncture studies of sodium-retaining animal models of the nephrotic syndrome have demonstrated pronounced $NaCl$ reabsorption in the distal nephron and TAL.^{395,396} The proteinuric kidney of a rat model of unilateral nephrotic syndrome has an enhanced Na^+ reabsorption in the CDs³⁹⁷ and a diminished response to ANP.³⁹⁸ Hyperaldosteronism reinforces $NaCl$ reabsorption at these sites. Renin and aldosterone levels are highly variable in patients with the nephrotic syndrome.³⁹⁹ Hypoalbuminemia reduces the binding of furosemide to plasma proteins and thereby enlarges its volume of distribution.⁴⁰⁰ Whereas one study has reported that premixing furosemide with albumin in the syringe prior to intravenous injection enhances the diuresis of patients with the nephrotic syndrome,¹⁰⁵ this finding

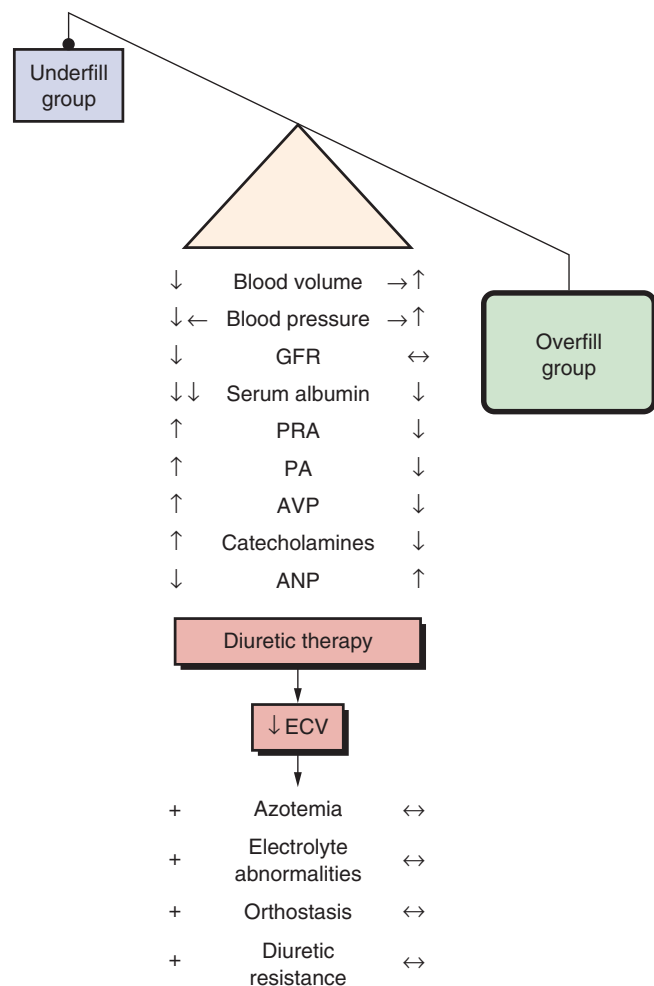


Fig. 50.17 Comparison of clinical and biochemical characteristics and responses in patients with nephrotic syndrome and underfill edema versus overfill edema. ANP, Atrial natriuretic peptide; AVP, arginine vasopressin; ↓ECV, decrease in extracellular fluid volume; GFR, glomerular filtration rate; PA, plasma aldosterone; PRA, plasma renin activity; ↑, increase; ↓, decrease; ↔, no effect; +, potential side effect. (Modified from Schrier RW, Fassett RG. A critique of the overfill hypothesis of sodium and water retention in the nephrotic syndrome. *Kidney Int.* 1998;53:1111–1117.)

has not been confirmed.^{106,401,402} Indeed, two studies have shown that patients with a serum albumin level of 2 g/dL can deliver normal quantities of furosemide into the urine.^{400,403} Iso-oncotic plasma volume expansion with albumin in patients with the nephrotic syndrome fails to induce negative NaCl balance⁴⁰⁴ or enhance the response to furosemide⁴⁰¹ and so is not generally recommended for the treatment of resistant nephrotic syndrome.^{401,405} One study has found that a fractional sodium excretion of 0.2% differentiates volume-contracted from volume-expanded nephrotic syndrome in children; treatment with diuretics alone has proven effective and safe for the volume-expanded group.⁴⁰⁶

A more logical approach to diuretic resistance is to limit albuminuria with an ACE inhibitor, ARB, or both, which also may combat the associated coagulopathy, dyslipidemia, edema, and progressive loss of renal function. The addition of a loop diuretic to an ACE inhibitor or an ARB reduces proteinuria further but increases the serum creatinine concentration.⁴⁰⁷

The tubular secretion of CAIs¹⁸ and loop diuretics¹²⁵ by the S2 segment of the proximal tubule depends on albumin. However, in the rabbit, the uptake of loop diuretics into the S1 portion of the proximal tubule, where furosemide is inactivated by glucuronidation, is inhibited by albumin (see Fig. 50.7).^{126,408} Albumin infusion into nephrotic patients does indeed increase renal furosemide excretion, whereas hypoalbuminemia enhances its metabolic clearance.⁴⁰⁹

The interaction of furosemide with its target in the lumen of the TAL is restricted by binding to filtered albumin.⁴¹⁰ Addition of albumin to the tubular perfusate of the loop of Henle attenuates the response to perfused furosemide because of binding to albumin and is reversed by coparfusion with warfarin, which displaces it from its albumin-binding site.⁴¹¹ However, Agarwal and colleagues have found that displacing furosemide from albumin via the coadministration of sulfisoxazole did not affect natriuresis in patients with the nephrotic syndrome.⁴¹² This study's results are not definitive, however, because subjects did not have diuretic resistance.

Animal studies have demonstrated five mechanisms that could impair the responsiveness to loop diuretics in patients with the nephrotic syndrome: (1, 2) decreased delivery and/or decreased tubular secretion of the diuretic; (3) increased renal metabolism; (4) decreased blockade by the diuretic; and (5) increased NaCl reabsorption by other nephron segments. Clinical studies have confirmed that nephrotic patients have an impaired tubular response to loop diuretics.

Nephrotic edema is best managed with dietary salt and fluid restriction. Most patients show an initial response to a loop diuretic when it is required. Spironolactone or eplerenone is effective in some patients.³⁹³ Decreasing renal function²⁹² or the administration of indomethacin²⁸¹ causes marked resistance to loop diuretics in these patients. The combination of a thiazide diuretic with furosemide dissipates edema but at the expense of marked kaliuresis.⁴¹³ Studies have suggested that primary sodium retention in nephrotic syndrome is related to the presence of plasma proteinases in nephrotic urine, such as plasmin, which can activate ENaC.^{414,415} A recent study has shown that this mechanism is also present in patients with proteinuric CKD.⁴¹⁶ This would provide a rationale for the ENaC blockers amiloride or triamterene, but requires clinical study.

IDIOPATHIC EDEMA

Idiopathic edema predominantly affects women. It causes fluctuating salt retention and edema, exacerbated by orthostasis.⁴¹⁷ The effects of diuretic withdrawal during controlled salt intake were studied in 10 such patients.⁴¹⁸ Although their body weight increased by 0.5 to 5.0 kg within 2 to 8 days, 7 returned to their original weight by 3 weeks, without the reinstitution of diuretic therapy. The investigators concluded that diuretic abuse could cause idiopathic edema. However, this conclusion has been challenged.⁴¹⁹ Remarkably, 83% of habitual furosemide abusers who consume high doses over prolonged periods demonstrate medullary nephrocalcinosis and tubulointerstitial fibrosis.⁴²⁰ Patients with idiopathic edema are best treated with salt restriction.

NONEDEMATOUS CONDITIONS

HYPERTENSION

Hypertension is discussed in Chapters 46 to 49.

ACUTE KIDNEY INJURY

A review of 11 randomized trials of loop diuretics or mannitol for prophylaxis or treatment of established AKI found no benefit.⁴²¹ Diuretics can be used to convert oliguric to nonoliguric AKI. One study involving 58 patients has found that sustained diuresis can be provoked in most patients given 1 g furosemide orally three times daily, but this very large dose produced deafness in two patients, which was permanent in one,⁴²² and therefore cannot be recommended. Older observational and randomized studies have indicated that furosemide does not improve the prognosis of AKI,^{423–425} and some have suggested that diuretics worsen the prognosis.⁴²⁶ Evidence of adverse effects of ECV overload in critically ill patients has been growing. The Fluid and Catheter Treatment Trial was a randomized trial comparing treatment strategies in patients with respiratory distress syndrome. A posthoc analysis of patients in this trial in whom AKI developed has suggested that patients randomly assigned to lower central venous pressure (CVP) targets exhibited lower mortality. These patients received substantially higher diuretic doses than those randomly assigned to higher CVP targets. Because this approach eliminated the confounding by indication that has complicated other studies, it provides substantial evidence that loop diuretics can be used safely in this population.⁴²⁷ Furosemide can reduce the need for dialysis by diminishing hyperkalemia, acidosis, or fluid overload.⁴²¹ One protocol is to give 40 mg of furosemide, 1 mg of bumetanide, or 25 mg of torsemide intravenously and to double the dose every 60 minutes if there is no response, up to a total daily dose of 1 g of furosemide or the equivalent.⁴²¹ Bumetanide and torsemide are metabolized by the liver and so may be preferred to furosemide, which is metabolized by the kidney and therefore accumulates to a greater degree in patients with renal insufficiency (see Fig. 50.6).

CHRONIC KIDNEY DISEASE

In subjects who are in balance, the fractional reabsorption of NaCl and fluid by the renal tubules is reduced in proportion to the fall in the GFR. The renal clearance of loop diuretics falls in parallel with the GFR because of a decreased renal

mass and the accumulation of organic acids that compete for proximal secretion.⁴²⁸ Thus, although the maximal increase in fractional excretion of Na⁺ produced by furosemide is maintained quite well in CKD,^{5,429,430} the absolute response to diuretics is limited by reductions in the absolute rate of NaCl reabsorption and in delivery of the diuretic to its target (see Fig. 50.15). Although CKD decreases proximal reabsorption, there is enhanced fractional reabsorption in the loop segment, distal tubule, and CDs,⁴³¹ with a relative increase of threefold to fourfold per residual nephron in the expression of the NKCC2 in the TAL and the NCC in the DCT.⁴³²

Torsemide has the greatest oral bioavailability in CKD.¹¹⁷ For refractory cases, a loop diuretic infusion (e.g., bumetanide, 1 mg/hour for 12 hours) produces a greater natriuresis and less myalgia than two bolus injections.²⁶⁸ Thiazides when used alone are considered relatively ineffective in patients with creatinine clearance below 35 mL/min/1.73 m², although this notion has been challenged.⁴³³ When used in combination with a loop diuretic that increases NaCl delivery and reabsorption at the distal tubule, larger doses of thiazides are effective in patients with moderate azotemia, although at the cost of a sharp further rise in the serum creatinine and blood urea concentrations and a high incidence of hypokalemia and electrolyte disorders.³¹¹ Moreover, high plasma levels of furosemide can cause ototoxicity.⁴³⁴ Therefore, care should be taken not to exceed the ceiling dose (Table 50.1).^{429,430} Epidemiologic studies have correlated diuretic use with end-stage CKD,⁴³⁵ but this is likely an epiphenomenon.⁴³⁶ In fact, continuing diuretics in patients undergoing dialysis who have residual renal function was associated with lower interdialytic weight gain, less hyperkalemia, and lower cardiac-specific mortality.⁴³⁷ In CKD patients using diuretics, the presence of hyponatremia indicated volume overload or depletion and was associated with the earlier initiation of renal replacement therapy.⁴³⁸ (See Wilcox⁴ and Sica and Gehr⁵⁸⁴ for further reading on diuretics in CKD.)

RENAL TUBULAR ACIDOSIS

Furosemide increases the distal delivery of NaCl and fluid and stimulates aldosterone secretion and phosphate elimination, which enhance acid elimination.⁴³⁹ In addition, a direct

Table 50.1 Ceiling Doses of Loop Diuretics (in mL)^a

Condition	Furosemide		Bumetanide, IV or PO	Torsemide, IV or PO
	IV	PO		
Chronic renal insufficiency:				
• Moderate (GFR = 20–50 mL/min/1.73 m ²)	80–160	160	6	50
• Severe (GFR <20 mL/min/1.73 m ²)	200	240	10	100
Nephrotic syndrome with normal GFR	120	240	3	50
Cirrhosis with normal GFR	40–80	80–160	1	20
Heart failure with normal GFR	40–80	80–160	1	20

^aFrom Skott P, Hommel E, Bruun NE, et al: The acute effect of acetazolamide on glomerular filtration rate and proximal tubular reabsorption of sodium and water in normal man. *Scand J Clin Lab Invest.* 1989;49:583–587; Schwartz GJ: Physiology and molecular biology of renal carbonic anhydrase. *J Nephrol.* 2002;15(Suppl 5):S61–S74; Kaunisto K, Parkkila S, Rajaniemi H, et al: Carbonic anhydrase XIV: luminal expression suggests key role in renal acidification. *Kidney Int.* 2002;61:2111–2118.

GFR, Glomerular filtration rate; IV, intravenous; PO, oral.

effect of both furosemide and thiazide on distal acidification with increased abundance of the H⁺-ATPase B1 subunit has been demonstrated.⁴⁴⁰ Hence, furosemide can be used in patients with hyperkalemic (type IV) renal tubular acidosis to increase renal acid excretion.^{441,442} Because hyperkalemic renal tubular acidosis is usually due to hypoaldosteronism, mineralocorticoid therapy is also often indicated.⁴⁴³

HYPERCALCEMIA

Ca²⁺ excretion is increased by osmotic or loop diuretics but decreased by thiazides and distal agents. Hypercalcemia activates the Ca²⁺-sensing receptor^{444,445} that inhibits fluid and NaCl reabsorption in the TAL and impairs renal concentration. The ensuing ECV depletion further limits Ca²⁺ excretion by reducing the GFR and enhancing proximal fluid and Ca²⁺ reabsorption. Therefore, the initial therapy for hypercalcemia is volume expansion with saline, with or without bisphosphonates or steroids, depending on the cause. Loop diuretics may help prevent or treat fluid overload, but there is little evidence to support a role in the treatment of hypercalcemia.⁴⁴⁶

NEPHROLITHIASIS

Thiazides reduce stone formation in hypercalciuric and even normocalciuric patients by reducing the excretion of Ca²⁺ and oxalate.⁴⁴⁷ Some patients continue to form stones and require additional citrate therapy.⁴⁴⁸ Ca²⁺ reabsorption can be enhanced further by the addition of amiloride⁴⁴⁹ or a low-salt diet. KHCO₃ produces a greater reduction in Ca²⁺ excretion than KCl when given with hydrochlorothiazide.⁴⁵⁰

OSTEOPOROSIS

Bone cells express an Na⁺-Cl⁻ cotransporter⁴⁵¹ that when blocked by a thiazide, enhances bone Ca²⁺ uptake.⁴⁵² Thiazides inhibit osteocalcin, an osteoblast-specific protein that retards bone formation,⁴⁵³ and directly stimulate the production of the osteoblast differentiation markers.^{454,455} They inhibit bone reabsorption⁴⁵⁶ and augment bone mineralization, independently of parathyroid hormone.⁴⁵⁷ Thus, thiazides may promote bone mineralization both by reducing renal Ca²⁺ excretion and through direct effects on bone. These biologic effects are supported by epidemiologic studies and clinical trial data. Thiazide therapy is associated with an increase in bone mineral density and a reduction in hip and pelvic fractures in older persons.^{458–460} In a placebo-controlled trial in postmenopausal women,⁴⁶¹ hydrochlorothiazide (50 mg/day) slowed cortical bone loss significantly. Surprisingly, despite having opposite effects on Ca²⁺ excretion, a thiazide and a loop diuretic both enhance bone formation in postmenopausal women, at least in the short term.⁴⁶² However, loop diuretics alone have been associated with hip bone loss in older men, increased risk of fractures in postmenopausal women, and increased risk of revision following primary total hip arthroplasty in men and women.^{463–465}

GITELMAN SYNDROME

In addition to potassium supplementation, potassium-sparing diuretics may be used in Gitelman syndrome to treat hypokalemia. Spironolactone (200–300 mg/day) was shown to be more effective than amiloride (10–30 mg/day).⁴⁶⁶ In a later study, the efficacies of indomethacin (75 mg/day), amiloride (30 mg/day), and eplerenone (150 mg/day) in

correcting hypokalemia were compared in a crossover trial.⁴⁶⁷ Although all three drugs significantly increased plasma potassium levels, indomethacin was most effective but also was associated with the most adverse effects.

DIABETES INSIPIDUS

Thiazides can reduce urine flow by up to 50% in patients with central or nephrogenic diabetes insipidus.^{468,469} This paradoxical effect is related to decreased GFR, enhanced water reabsorption in the proximal and distal nephron,^{470,471} and an increase in papillary osmolarity leading to distal water reabsorption. A small placebo-controlled crossover trial and an animal study have shown that amiloride prevents lithium-induced polyuria.^{472,473} This effect is attributed to blockage by amiloride of the entry of lithium via ENaC in the principal cell, where it can downregulate the water channel AQP2 via glycogen synthase kinase 3.⁴⁷⁴

ADVERSE EFFECTS OF DIURETICS

FLUID AND ELECTROLYTE ABNORMALITIES

EXTRACELLULAR VOLUME DEPLETION AND AZOTEMIA

Diuretics normally do not decrease the GFR.^{83,475} However, a GFR decrease can be precipitated by vigorous diuresis in patients with CKD, severe edema, or cirrhosis and ascites. A rise in the ratio of blood urea nitrogen to creatinine suggests ECV depletion. This change can be ascribed to decreased renal urea clearance because of greater urea reabsorption in the distal nephron⁴⁷⁶ and to increased urea generation. The latter is due to greater uptake by the liver of arginine, which is metabolized by arginase.^{477–479} In addition, combining diuretics with ACE inhibitors and NSAIDs raises the risk of AKI.^{480,481}

HYPONATREMIA

This effect is relatively specific for thiazides, which inhibit urinary dilution, whereas loop diuretics inhibit urinary concentration and dilution.⁴⁸² Indeed, thiazides are 12-fold more likely than loop diuretics to cause hyponatremia.⁴⁸³ The mechanism of thiazide-induced hyponatremia has long remained elusive. Although one could postulate that the natriuretic effect causes plasma volume depletion with AVP release, patients with thiazide-induced hyponatremia actually gain weight.⁴⁸⁴ Clark and colleagues have shown that older age and thiazide diuretics are additive in impairing maximal free water excretion following a water load.⁴⁸⁵ Some 80% of thiazide-induced hyponatremia occurs in females,⁴⁸⁶ most of whom are older and with low body mass.^{487,488} Hyponatremia can develop during rechallenge with a thiazide.^{484,489,490} It often develops within the first 2 weeks of thiazide therapy.^{483,489} A recent study that combined phenotypic and pharmacogenetic data has integrated these previous observations into a more consistent pathophysiologic mechanism.⁴⁹¹ Patients with thiazide-induced hyponatremia displayed increased free water reabsorption independently of AVP and increased urinary PGE2 excretion (Fig. 50.18). Pharmacogenetic analysis has identified a variant in *SLCO2A1*, which encodes a prostaglandin transporter in the distal nephron. In vitro, the variant of *SLCO2A1* displayed loss of function. A reduced or absent

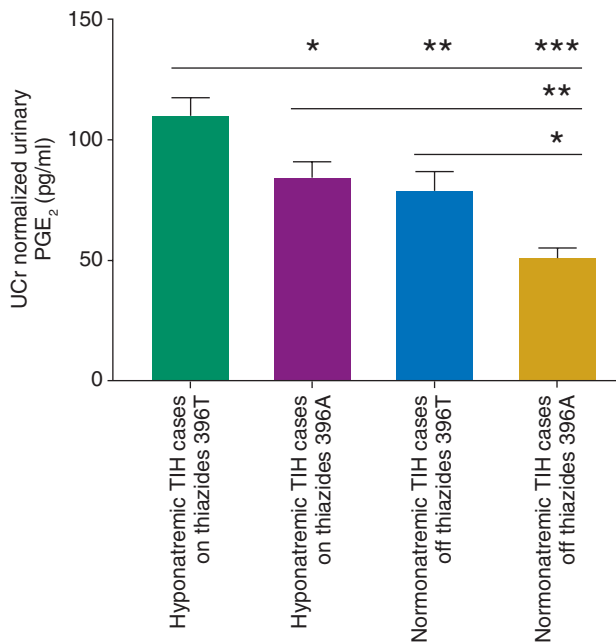


Fig. 50.18 Urinary prostaglandin E₂ (PGE₂) concentration in thiazide-induced hyponatremia (TIH) cases, compared with normonatremic thiazide- and non-thiazide-treated controls, stratified according to their genotype at position 396 of the prostaglandin transporter gene, *SLCO2A1*. TIH cases who carry at least one copy of the risk allele (396T) have significantly elevated urinary PGE₂ levels relative to those homozygous for 396A, whereas no such effect was observed in normonatremic controls. Data are represented as mean $\pm$ SEM. * $P < .05$; ** $P < .01$; *** $P < .001$. UCr, Urinary creatinine. (From Ware JS, Wain LV, Channavajjhala SK, et al. Phenotypic and pharmacogenetic evaluation of patients with thiazide-induced hyponatremia. *J Clin Invest*. 2017;127(9):3367–3374.)

prostaglandin transporter in the apical plasma membrane allows PGE₂ to stimulate EP₄ receptors, which can subsequently insert AQP2 water channels, increasing water reabsorption (Fig. 50.19).

Mild hyponatremia can be treated by withdrawal of diuretics, restriction of the daily intake of free water to 1.0 to 1.5 L, increasing solute intake, restoration of any K⁺ and Mg²⁺ losses, and replenishment of NaCl if the patient is clearly volume-depleted.^{483,492}

It is becoming increasingly clear that even mild and chronic hyponatremia can produce symptoms.⁴⁹³ Therefore, the development of diuretic-induced hyponatremia should probably be regarded as a contraindication to continue using the agent responsible. Despite the importance of diuretic-induced hyponatremia (and hypokalemia; see later) for individual cases, these side effects are relatively mild on a population basis and should therefore not discourage prescription of these effective drugs. For example, in a study of 3000 patients starting antihypertensive monotherapy, serum sodium and potassium values were only marginally lower in thiazide users, with more than 90% of patients maintaining normal levels.⁴⁹⁴

HYPOKALEMIA

Four mechanisms have been identified that increase renal K⁺ elimination during therapy with thiazides or loop diuretics (Fig. 50.20)—increased tubular flow, secretion of AVP, secretion of aldosterone, and alkalosis. Flow-dependent K⁺ secretion by the distal nephron provides a universal mechanism for increased K⁺ secretion in response to diuretics that act more proximally.¹⁷² The basal and flow-dependent components of K⁺ secretion have been shown to be largely mediated by distinct channels. Whereas basal secretion traverses ROMK channels, flow-dependent K⁺ secretion is mediated

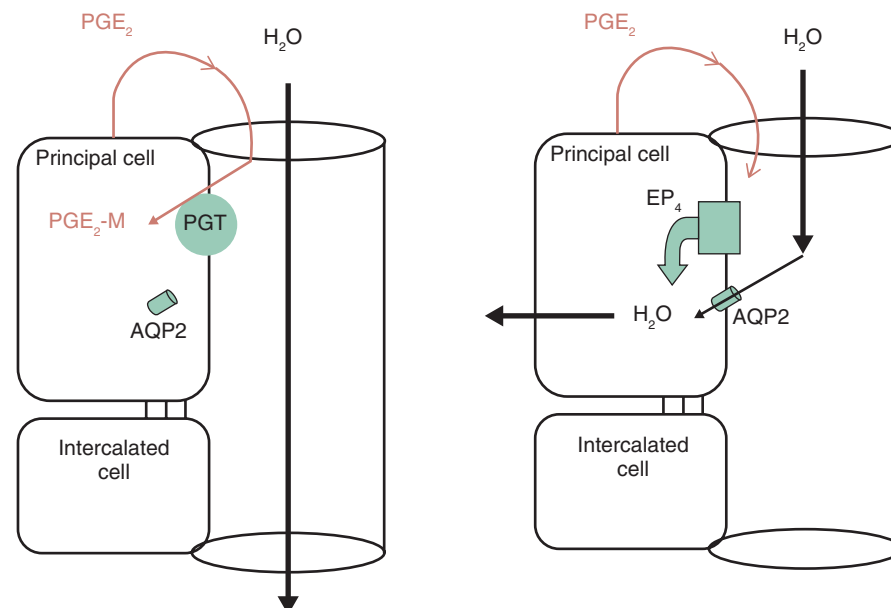


Fig. 50.19 Hypothesis for the role of *SLCO2A1* in contributing to TIH in individuals carrying the *SLCO2A1* A396T variant. *Left*, Under low ADH conditions, apical PGT in the renal collecting duct scavenges PGE₂ from the lumen, resulting in AQP2 internalization and minimal osmotic water reabsorption. *Right*: With reduced or absent apical PGT, PGE₂ reaching the lumen is able to stimulate apical EP₄ receptors, resulting in the insertion of AQP2 and osmotic water reabsorption. ADH, Antidiuretic hormone; AQP2, aquaporin-2; PGE₂, prostaglandin E₂; PGT, phosphate glucuronyl transferase; TIH, thiazide-induced hyponatremia.

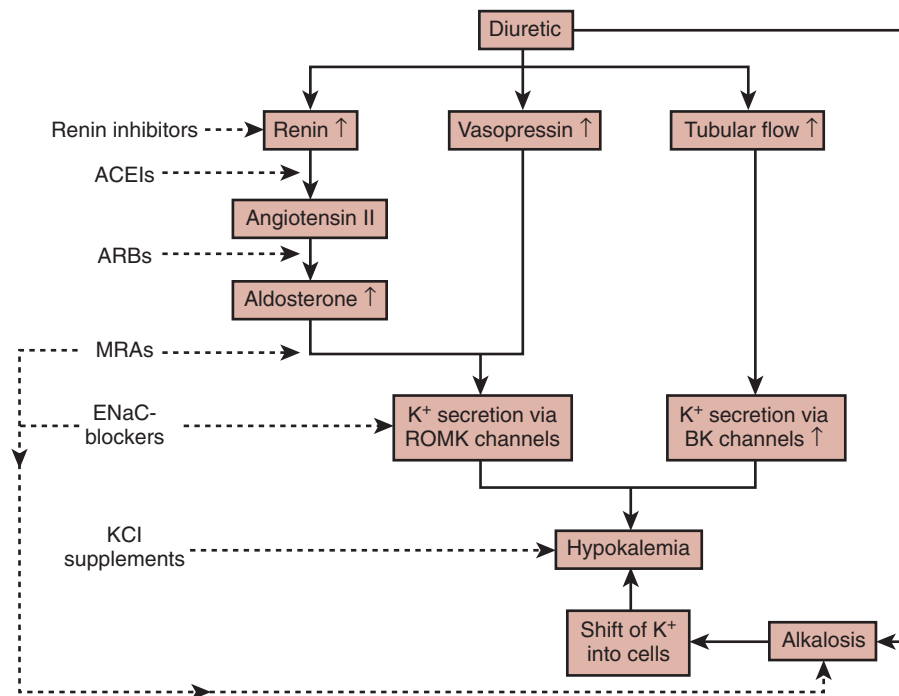


Fig. 50.20 Diagrammatic representation of mechanisms that increase ($\uparrow$) K^+ excretion by the collecting ducts or partition K^+ into cells during therapy with a thiazide or loop diuretic, and strategies for prevention or treatment (dashed lines) with direct renin inhibitors, angiotensin-converting enzyme inhibitors (ACEIs), angiotensin receptor blockers (ARBs), mineralocorticoid receptor antagonists (MRAs; e.g., spironolactone), epithelial Na^+ channel (ENaC) blockers (amiloride or triamterene), or KCl supplements. Diuretics stimulate the renin-angiotensin-aldosterone system (RAAS), increase distal tubule flow and release of arginine vasopressin, and generate a metabolic alkalosis, all of which enhance K^+ secretion in the collecting ducts. Two distinct mechanisms are responsible for K^+ secretion in the collecting ducts, one mediated by the renal outer medulla K^+ (ROMK) channels, and the other by big K^+ (BK) channels.

by calcium-activated maxi or big K^+ (BK) channels.^{495,496} Normally, a flow-dependent rise in K^+ secretion during increased water intake is offset by a drop in AVP concentration that diminishes distal K^+ secretion. Diuretic therapy, however, is unusual because it combines increased distal tubule flow with maintained or increased AVP release due to nonosmotic stimulation, the second mechanism contributing to kaliuresis. Indeed, inhibition of AVP release in normal subjects undergoing a furosemide diuresis inhibits the kaliuresis.²⁷⁷ Nonosmotic AVP release is common in edematous subjects.⁴⁹² Therefore, enhanced release of AVP and increased distal tubule fluid delivery combine to promote ongoing K^+ losses during diuretic therapy for edema. Diuretic-induced aldosterone secretion also promotes distal K^+ secretion.^{172,295,497} The effects of flow and aldosterone on distal K^+ secretion are normally counterbalanced during changes in salt intake, just as the effects of flow and AVP induced by changes in water intake (see Fig. 50.20). Diuretic treatment, however, uncouples the two because it enhances the secretion of aldosterone and AVP but increases distal flow, thereby accounting for the particular importance of aldosterone and AVP in promoting K^+ loss with diuretics.²⁷⁷ Finally, diuretic-induced alkalosis enhances the distal secretion of K^+ .¹⁷²

The serum potassium concentration in patients not receiving KCl supplements falls by an average of 0.3 mmol/L with furosemide and 0.6 mmol/L with thiazides.⁴⁹⁸ This 20% fall in serum potassium level is accompanied by a fall in total body K^+ that averages less than 5%.⁴⁹⁸ Moreover, in normal subjects receiving daily doses of loop diuretics, there is no

detectable change in K^+ balance, despite a reproducible fall in serum potassium level of about 0.5 mmol/L.²⁷⁷ This finding implies that the primary cause of hypokalemia during diuretic administration is a redistribution of K^+ into cells, likely related to the accompanying metabolic alkalosis.^{70,123}

Mild, diuretic-induced hypokalemia (serum potassium, 3.0–3.5 mmol/L) increases the frequency of ventricular ectopy.⁴⁹⁹ Some investigators have shown that thiazide-induced hypokalemia does not pose a risk of clinically significant cardiac dysrhythmia, even in large populations of hypertensive patients.^{381,500} In contrast, others have reported a dose-dependent risk of cardiac arrest in patients receiving thiazides that is prevented by therapy with a K^+ -sparing diuretic such as amiloride.⁵⁰¹ In one study in rats, thiazide-induced hypokalemia was associated with hypomagnesemia, hypertriglyceridemia, insulin resistance, hyperaldosteronism, and renal injury, all of which were absent in rats with a similar degree of hypokalemia due to a dietary potassium depletion.⁵⁰² Although of concern, these data may be species-specific and may also be ascribed to the dose given (10 mg/kg).⁵⁰³ In a large population-based study of older adults, chlorthalidone was shown to be as effective in reducing cardiovascular outcomes as hydrochlorothiazide but more often resulted in hospitalization for hypokalemia or hyponatremia.⁵⁰⁴

Adverse effects of hypokalemia are clearly important in certain cases:

1. Severe hypokalemia (serum potassium < 3.0 mmol/L) requires treatment because it is associated with a doubling

- of serious ventricular dysrhythmias, muscular weakness, and rhabdomyolysis.⁵⁰⁵
2. Mild hypokalemia can precipitate dangerous dysrhythmias in patients with cardiac dysfunction due to left ventricular hypertrophy, coronary ischemia, HF, prolonged QT interval, anoxia, or ischemia, as well as in patients with known dysrhythmias.
 3. Hypokalemia enhances the toxicity of cardiac glycosides by diminishing the renal tubule secretion of digoxin and by enhancing its binding to cardiac Na⁺-K⁺-ATPase, thereby exaggerating its actions on the heart.
 4. Hypokalemia stimulates renal ammoniogenesis. This effect is dangerous for patients with cirrhosis and ascites who are prone to the development of hepatic encephalopathy due to hyperammonemia. Moreover, the accompanying diuretic-induced alkalosis partitions ammonia into the brain.
 5. Catecholamines partition K⁺ into cells and lower serum potassium levels. Myocardial infarction provokes sufficient catecholamine release to lower serum potassium levels by approximately 0.5 mmol/L, which is potentiated in patients who have received prior thiazide therapy.⁵⁰⁶
 6. Hypokalemia impairs insulin release and predisposes to hyperglycemia.⁵⁰⁷
 7. Hypokalemia limits the antihypertensive action of thiazides.⁵⁰⁸ In a placebo-controlled study of hypokalemic subjects receiving thiazide diuretics, coadministration of KCl that restored serum potassium levels also reduced BP significantly.⁵⁰⁸

Therefore, it is prudent to prevent even mild degrees of hypokalemia. Hypokalemia can be prevented by increasing the intake of K⁺ with Cl⁻ (see Fig. 50.20) but doing so often requires 40 to 80 mmol daily. Moreover, in the presence of alkalosis, hyperaldosteronism, or Mg²⁺ depletion, hypokalemia is quite unresponsive to dietary KCl. A more effective, convenient, and predictable strategy is to prescribe a combined therapy with a distal K⁺-sparing agent such as amiloride or triamterene, which maintains serum potassium levels during short- or long-term hydrochlorothiazide therapy (see Fig. 50.20).^{509,510} It also prevents diuretic-induced alkalosis and provides further natriuresis and antihypertensive efficacy. An alternative strategy is to administer an ACE inhibitor, ARB, or MRA to counter angiotensin II-induced hyperaldosteronism, which would promote distal K⁺ secretion. The fall in BP and the beneficial cardiovascular actions of these agents are clearly a further advantage. Finally, a small study has shown that combining torsemide with hydrochlorothiazide leads to a synergistic increase in sodium excretion but, surprisingly, reduces urinary potassium and magnesium losses in comparison with hydrochlorothiazide alone,⁵¹¹ possibly owing to the antialdosteronic effects of torsemide.⁵¹²

HYPERKALEMIA

Diuretics acting in the ASDN decrease K⁺ secretion and predispose to hyperkalemia.^{497,513} As noted previously, this complication was observed more frequently after increased use of distal K⁺-sparing diuretics in Canada,²¹¹ although a later report from Scotland was not able to reproduce these findings.⁵¹⁴ The risk of hyperkalemia may rise, especially when spironolactone is combined with other drugs that interfere

with renal potassium secretion, such as ACE inhibitors or the antibiotic trimethoprim-sulfamethoxazole.^{515,516}

HYPOMAGNESEMIA

Loop diuretics inhibit Mg²⁺ reabsorption in the TAL^{78,517} by reducing the transepithelial voltage that drives Mg²⁺ and Ca²⁺ paracellularly (see Fig. 50.4).⁵¹⁸ Thiazides first enhance Mg²⁺ uptake in the DCT but, during prolonged therapy, there is enhanced renal Mg²⁺ excretion.¹⁵⁶ Although this greater excretion has been attributed to a fall in cellular [Na⁺] that stimulates basolateral Na⁺-Mg²⁺ exchange,⁵¹⁸ later molecular data have suggested that long-term thiazide use leads to the downregulation of TRPM6, the predominant apical Mg²⁺ channel of the distal nephron.^{153,519} Distal K⁺-sparing agents and spironolactone diminish Mg²⁺ excretion. During prolonged therapy with thiazides and loop diuretics, serum Mg²⁺ concentration falls by 5% to 10%. Diuretic-induced hyponatremia and hypokalemia cannot be reversed fully until any Mg²⁺ deficit is replaced.⁵²⁰ Mg²⁺-depleted rats secrete K⁺ inappropriately into the distal tubule, independently of aldosterone,⁵²¹ possibly because a decrease in intracellular magnesium releases the magnesium-mediated inhibition of ROMK channels.⁵²²

HYPERCALCEMIA

Thiazides increase the serum concentrations of total and ionized calcium but rarely result in frank hypercalcemia. During established thiazide treatment, parathyroid hormone concentrations are inversely related to ionized serum calcium.⁵²³ The increased serum calcium can be ascribed primarily to enhanced Ca²⁺ reabsorption (the mechanism discussed previously). Persistent hypercalcemia should prompt a search for a specific cause—for example, an adenoma of the parathyroid glands.^{523,524} A recent epidemiologic study from the United States has found an annual incidence of thiazide-associated hypercalcemia of 12/100,000.⁵²⁵ Of the 221 subjects who were identified with thiazide-associated hypercalcemia, 53 (24%) had primary hyperparathyroidism.

ACID-BASE CHANGES

Metabolic alkalosis induced by thiazides or loop diuretics is an important adverse factor in patients with hepatic cirrhosis and ascites, in whom the alkalosis may provoke hepatic coma by partitioning ammonia into the brain, and in those with underlying pulmonary insufficiency, in whom the alkalosis diminishes ventilation.⁵²⁶ The generation of metabolic alkalosis with loop diuretics results from contraction of the extracellular HCO₃⁻ space by the excretion of a relatively HCO₃⁻-free urine.^{70,527} The maintenance of metabolic alkalosis involves increased net acid excretion in response to hypokalemia-induced ammoniogenesis and mineralocorticoid excess during continued Na⁺ delivery and reabsorption at the distal nephron sites of H⁺ secretion.⁵²⁸ Diuretic-induced metabolic alkalosis is best managed with the administration of KCl, but a distal K⁺-sparing diuretic⁵²⁹ or, occasionally, a CA inhibitor, should be considered.¹⁹ Metabolic alkalosis impairs the natriuretic response to loop diuretics¹²³ (see Fig. 50.11) and may thereby contribute to diuretic resistance.

CA inhibitors produce metabolic acidosis. Spironolactone, eplerenone, amiloride, and triamterene can cause hyperkalemic metabolic acidosis, especially in older patients, patients with renal impairment, and those receiving KCl supplements.

METABOLIC ABNORMALITIES

HYPERGLYCEMIA

Diuretic therapy, especially with thiazides, impairs carbohydrate tolerance and occasionally precipitates diabetes mellitus.^{530–532} This increase is attributed to decreased hepatic utilization of glucose.⁵³³ Hyperglycemia persists⁵³⁴ but is reversed rapidly after diuretic discontinuation, even after 14 years of thiazide therapy.⁵³⁵ The increase in blood glucose levels provoked by thiazides is worsened by concurrent β -adrenergic receptor blockade.⁵³³ A meta-analysis has shown that thiazide diuretics increase fasting blood glucose levels by approximately 30.4 mg/dL (1.69 mmol/L).⁵³⁶

Thiazides impair glucose uptake into muscle^{537,538} and liver.⁵³³ This effect is more pronounced during initiation of therapy. It has been ascribed to a diuretic-induced reduction in cardiac output with reflex activation of the SNS and catecholamine secretion, which lead to reductions in hepatic glucose uptake, muscle blood flow, and muscle glucose uptake (Fig. 50.21). During sustained thiazide therapy, there is decreased insulin release, which can be corrected by reversal of hypokalemia with KCl,⁵³⁹ hypomagnesemia with magnesium oxide,⁵⁴⁰ or administration of amiloride or spironolactone.

Experimental K^+ deficiency causes glucose intolerance and impairs insulin secretion in the absence of diuretics.⁵⁴¹ Hydrochlorothiazide and ACE inhibitors have opposite effects on glucose disposal, which are attributed to their opposing effects on serum potassium levels.⁵³⁸ The increase in serum glucose levels that occurs with thiazide therapy is more pronounced in obese patients and correlates with a fall in intracellular $[K^+]$ or $[Mg^{2+}]$.⁵⁴² It can be prevented by reducing the thiazide dosage⁵⁰⁷ or by KCl replacement.⁵³⁹ Therefore, care should be taken to monitor blood glucose levels during thiazide therapy, particularly in obese or diabetic patients, and to prevent hypokalemia.

Thiazide-induced hyperglycemia should be anticipated and prevented. Measures include coadministration of a distal K^+ -sparing diuretic, ACE inhibitor, or ARB, prescribing extra KCl, or reducing the thiazide dosage.^{510,543} Thiazide diuretics are not contraindicated in patients with diabetes mellitus. Indeed, diuretics produce an even greater reduction in the absolute risk for cardiovascular events in hypertensive patients who are diabetic.⁵⁴⁴ Therefore, diabetes mellitus is an indication for close surveillance and attention to the coadministration of agents designed to prevent hypokalemia (see Fig. 50.21).

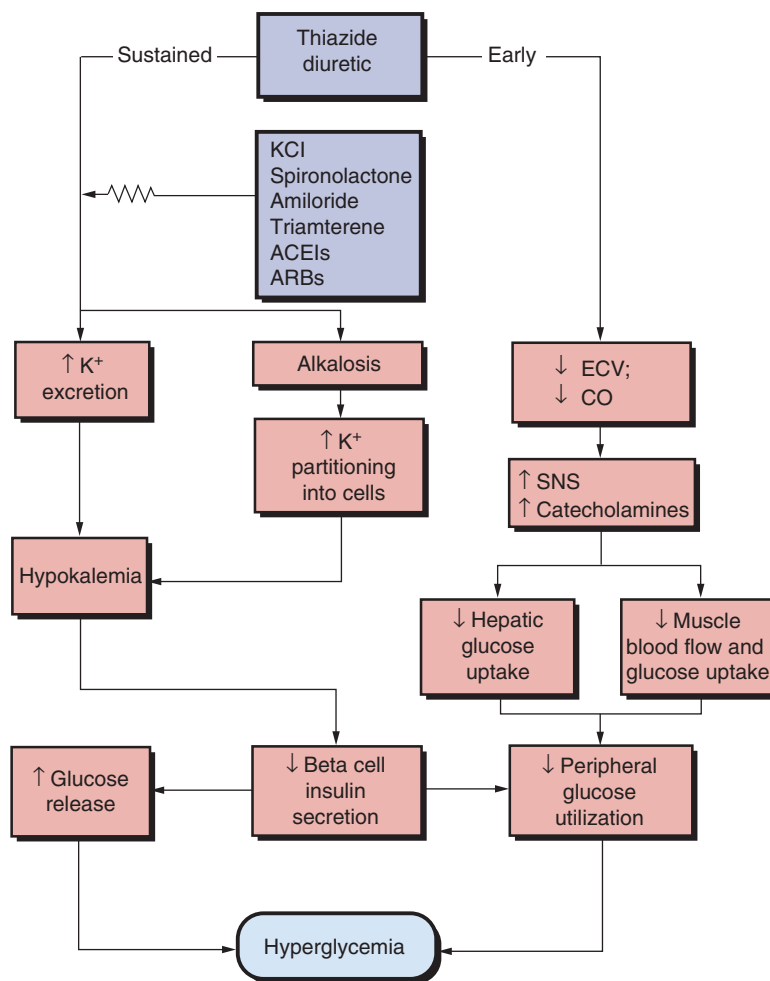


Fig. 50.21 Hypothesis for the hyperglycemic actions of thiazide diuretics. ACEI, Angiotensin-converting enzyme inhibitor; ARB, angiotensin receptor blocker; ECV, extracellular fluid volume; SNS, sympathetic nervous system; CO, cardiac output; ↑, increase(d); ↓, decrease(d). (From Wilcox CS. Metabolic and adverse effects of diuretics. *Semin Nephrol.* 1999;19:557–568.)

HYPERLIPIDEMIA

Administration of loop diuretics or thiazides raises the plasma concentrations of total cholesterol, triglycerides, and low-density lipoprotein cholesterol, but reduces the concentration of high-density lipoprotein cholesterol. These adverse changes average 5% to 20% during initiation of therapy.⁵⁴⁵ The mechanism is uncertain; it may relate to ECV depletion because severe dietary NaCl restriction has similar metabolic effects,⁵⁴⁶ whereas increasing NaCl intake lowers serum cholesterol levels. Alternatively, it may relate to hypokalemia, which impairs insulin secretion.^{539,542} Importantly, most studies have shown that serum cholesterol levels return to baseline over 3 to 12 months of thiazide therapy.^{545,547} In fact, when combined with lifestyle management, 4 years of thiazide therapy for hypertension is associated with a modest improvement in the lipid profile.^{548,549}

HYPERURICEMIA

Prolonged thiazide therapy for hypertension increases the serum urate concentration by approximately 35%. Renal urate clearance falls because of competition for secretion between urate and the diuretic⁸¹ and ECV depletion–induced urate reabsorption (see Fig. 50.5).⁸² Hyperuricemia is dose-related and can lead to gout. A very long-term outcome analysis of 3693 patients detected no adverse effects of diuretic-induced hyperuricemia in hypertensive subjects who did not have gout.⁵⁵⁰ However, later studies have correlated raised serum urate levels with a higher cardiovascular death rate.⁵⁵¹ One possibility is to combine diuretic treatment with the ARB losartan, which has weak uricosuric properties.⁵⁵²

OTHER ADVERSE EFFECTS

IMPOTENCE

In the Medical Research Council trial involving 15,000 hypertensive subjects, impotence was much higher in those receiving a thiazide.⁵³⁰ In the Treatment of Mild Hypertension Study, erection problems were twice as high in subjects receiving a thiazide as in those who were not.⁵⁵³ A controlled trial has demonstrated the efficacy of sildenafil in reversing impotence in hypertensive patients receiving multiple antihypertensive agents, including diuretics.⁵⁵⁴

OTOTOXICITY

Loop diuretics can cause deafness that may occasionally be permanent.⁵⁵⁵ The risk is greater with ethacrynic acid than with other loop diuretics and is also greater when a loop diuretic is combined with another ototoxic drug (e.g., an aminoglycoside).⁵⁵⁶ It is especially common during high-dose bolus intravenous therapy in patients with renal failure, in whom plasma levels are increased, and in hypoalbuminemic subjects.⁵⁵⁷ In a crossover trial, no ototoxicity was noted in patients with severe HF when they were given an infusion of 250 to 2000 mg of furosemide over 8 hours, whereas reversible deafness occurred in 25% when the same dose was given as a bolus.³⁵⁴

HAZARDS IN PREGNANCY AND NEWBORNS

Diuretics do not prevent preeclampsia. They have little effect on perinatal mortality.⁵⁵⁸ Thiazide therapy can be continued

during pregnancy in patients whose hypertension has been controlled by these agents or used to treat pulmonary edema.⁵⁵⁹

Intensive therapy with loop diuretics for neonates with respiratory distress syndrome increases the prevalence of patent ductus arteriosus because of the increased generation of prostaglandins,⁵⁶⁰ cholelithiasis, secondary hyperparathyroidism, bone disease, and drug fever.⁵⁶¹ Prolonged furosemide therapy in preterm infants can cause renal calcification.⁵⁶² Diuretics can be transferred from the mother to the infant in breast milk,⁵⁶² in whom they can cause serious fluid and electrolyte abnormalities.⁵⁶³

VITAMIN B DEFICIENCY

Diuretics increase the excretion of water-soluble vitamins.⁵⁶⁴ Long-term diuretic therapy for HF reduces folate and vitamin B₁ (thiamine) levels⁵⁶⁵ and increases plasma homocysteine levels. Thiamine can improve left ventricular function in some patients with HF being treated with furosemide.⁵⁶⁶

DRUG ALLERGY

A reversible photosensitivity dermatitis occurs rarely during thiazide or furosemide therapy.⁵⁶⁷ High-dose furosemide in renal failure can cause bullous dermatitis.⁵⁶⁸ Diuretics may cause a more generalized dermatitis, sometimes with eosinophilia, purpura, or blood dyscrasia. Occasionally, they cause a necrotizing vasculitis or anaphylaxis.⁵⁶⁹ Acute allergic reactions to sulfonamides are mediated via immunoglobulin E, whereas delayed-onset hypersensitivities are mediated by antibodies to specific protein epitopes.⁵⁷⁰ Severe necrotizing pancreatitis is a rare complication.⁵⁷¹ Acute interstitial nephritis with fever, rash, and eosinophilia may develop abruptly some months after initiation of therapy with a thiazide or, less often, furosemide.^{572,573} Ethacrynic acid is chemically dissimilar from other loop diuretics and can be a substitute.

MALIGNANCY

Thiazide diuretics have been associated with both skin and lip cancers, possibly through their photosensitizing properties.^{574,575} Although previous epidemiologic studies have suggested associations between diuretic use and cancer (renal cell carcinoma and colon carcinoma), a 2011 meta-analysis has refuted the suggestion of a higher risk of cancer or cancer-related death with use of any antihypertensive drugs.⁵⁷⁶

ADVERSE DRUG INTERACTIONS

Hyperkalemia in patients receiving distal K⁺-sparing diuretics, spironolactone, or eplerenone can be precipitated by concurrent therapy with KCl, ACE inhibitors, ARBs, heparin, ketoconazole, trimethoprim, or pentamidine. Therefore, these drugs should not normally be prescribed in combination, especially in patients with impaired renal function or diabetes. ACE inhibitors increase the risk of severe hyperkalemia in patients with decompensated HF receiving spironolactone.^{577,578} Loop diuretics and aminoglycosides potentiate ototoxicity and nephrotoxicity.^{557,579} Diuretic-induced hypokalemia increases digitalis toxicity fourfold.⁵⁸⁰ Plasma lithium concentrations rise during loop diuretic therapy⁵⁸¹ because of increased proximal lithium reabsorption.⁵⁸² NSAIDs may impair the diuretic, natriuretic, antihypertensive, and venodilating responses to

diuretics and predispose to renal vasoconstriction and a drop in the GFR (see earlier discussion, “Adaptation to Diuretic Therapy”). Used together, indomethacin and triamterene may precipitate renal failure.⁵⁸³

 Complete reference list available at [ExpertConsult.com](https://www.expertconsult.com).

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BOARD REVIEW QUESTIONS

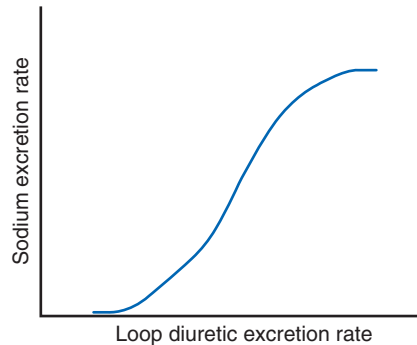
1. Which of the following is not a useful strategy for diuretic resistance?
- Dietary sodium restriction
 - Premixing furosemide with albumin prior to intravenous administration
 - Switching from oral to intravenous administration in equal dosing
 - Combining different types of diuretics

Answer: b

2. Which of the following mechanisms does not contribute to the tendency of loop diuretics to cause hypokalemia?
- Increased tubular flow
 - Increased secretion of vasopressin
 - Increased secretion of aldosterone
 - Increased secretion of prostaglandins

Answer: d

3. In the figure, the normal dose-response curve for loop diuretics is shown. How is this dose-response curve affected by chronic kidney disease (CKD) and heart failure (HF)?



- CKD and HF only cause a rightward shift of the curve.
- CKD and HF only cause a downward shift of the curve.
- CKD only causes a rightward shift of the curve, whereas HF causes both rightward and downward shifts of the curve.
- CKD causes both rightward and downward shifts of the curve, whereas HF only causes a rightward shift of the curve.

Answer: c